Pharmacodynamics

I. Definitions:

A. Drugs:

- **Chemical agents that interact with components of a biological system to alter the organism's function, used to treat, cure, prevent, or diagnose a disease or to promote well-being.**

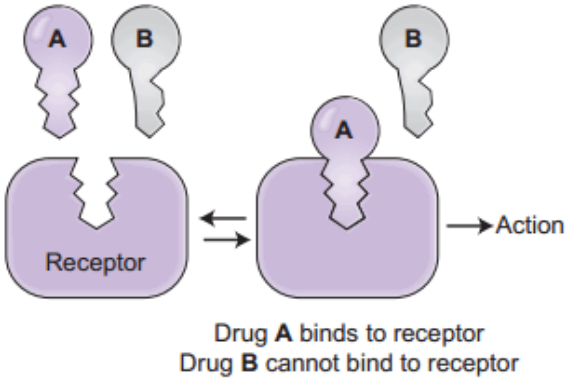
- Examples of such components (sites of drug action) are:
  - Enzymes
  - Ion channels
  - Neurotransmitter transport systems
  - Nucleic acids
  - **Receptors**

- Many drugs act by mimicking or inhibiting the interactions of endogenous mediators with their receptors

B. Receptors:

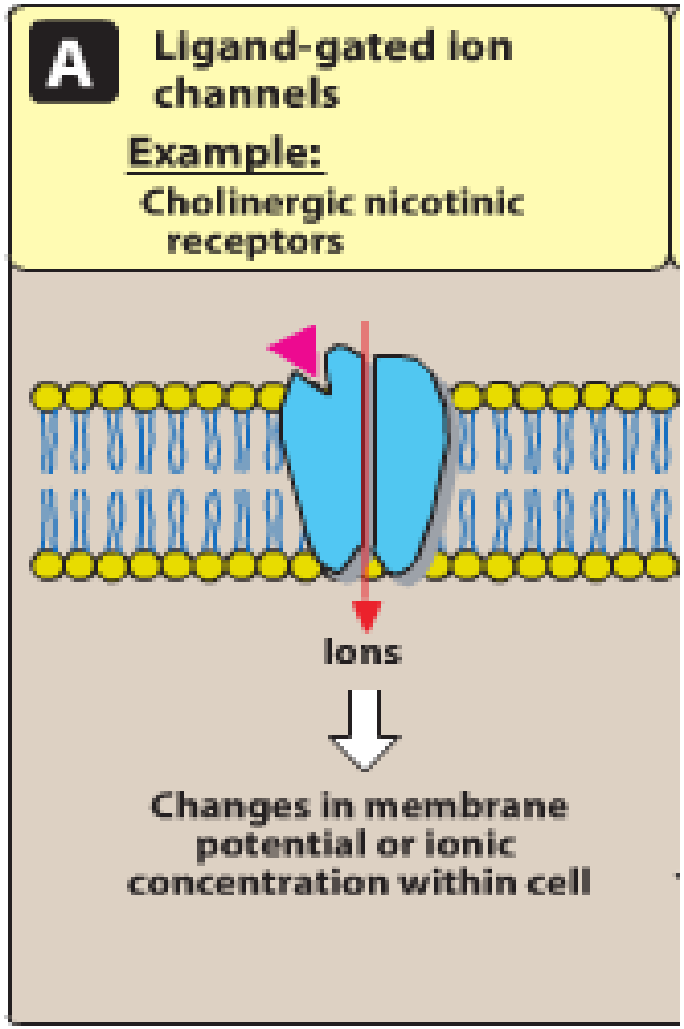
- **Regulatory proteins that interact with drugs or hormones and initiate a cellular response**

- **<u>4 types of receptors:</u>**
  1. Ion channels (Ligand-gated Ion Channels)
  2. G-protein coupled receptors
  3. Receptor-enzymes
  4. Cytosolic-nuclear receptors

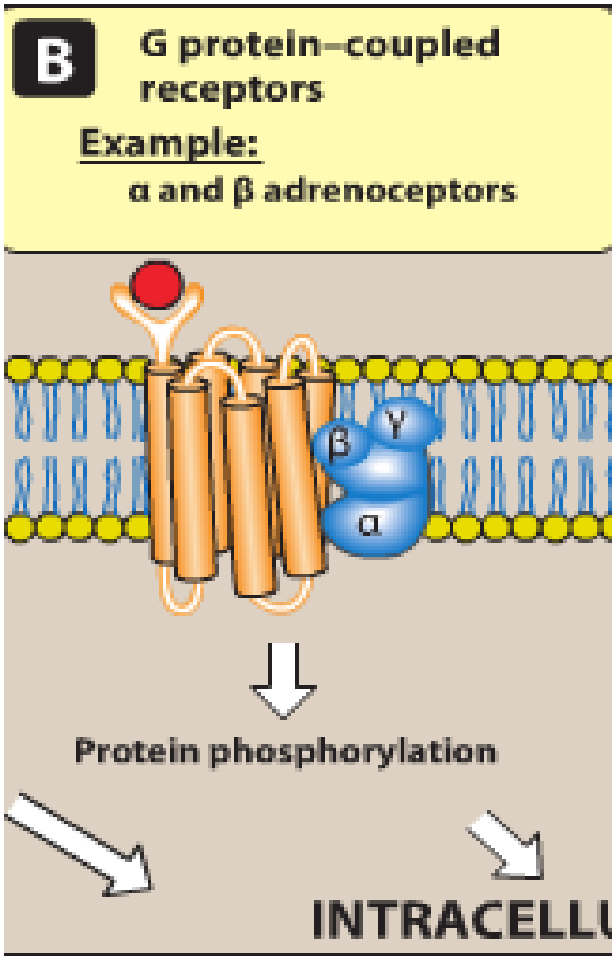


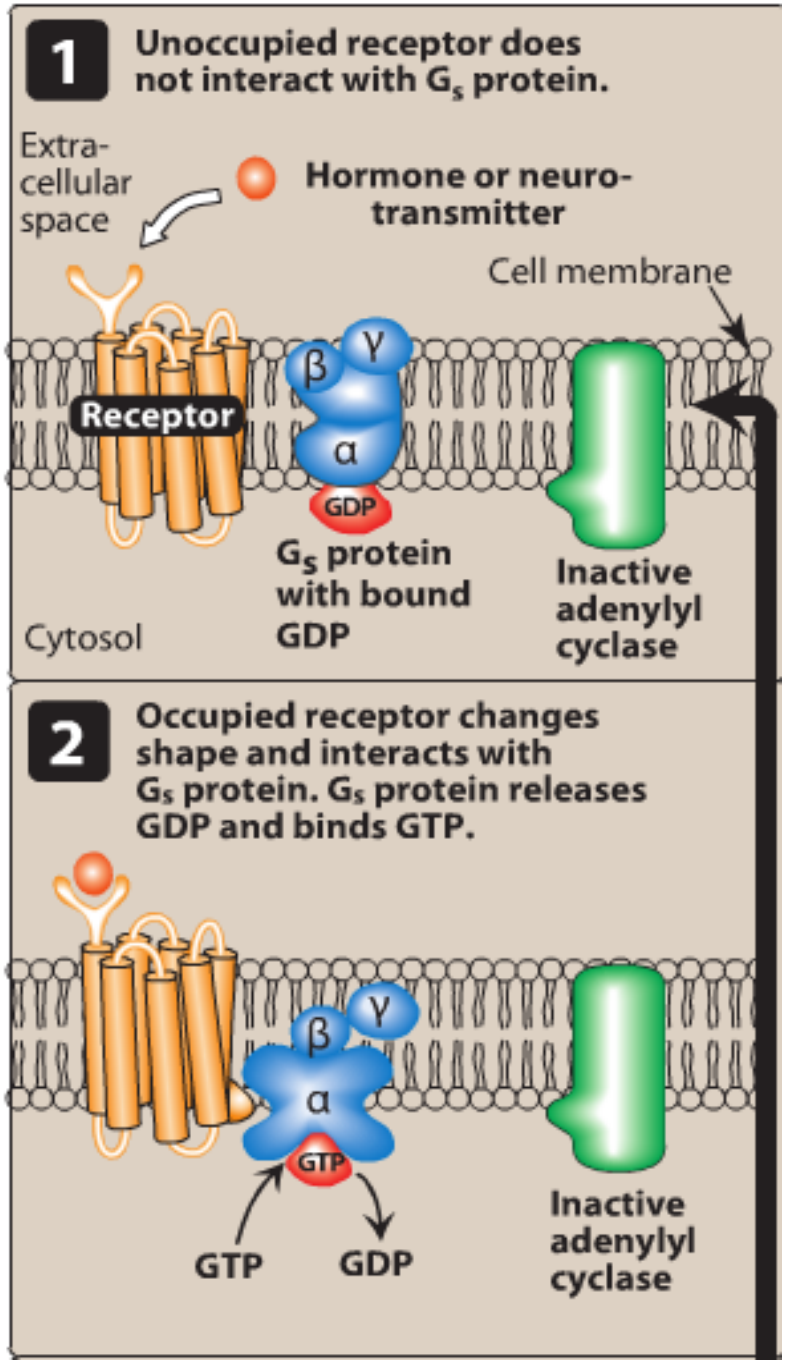
- **Act as transducer proteins**
  - Receptor-effector signal transduction
  - Post-receptor signal transduction provides for amplification of the signal

II. Types of Receptors
1. Ligand-gated Ion Channels

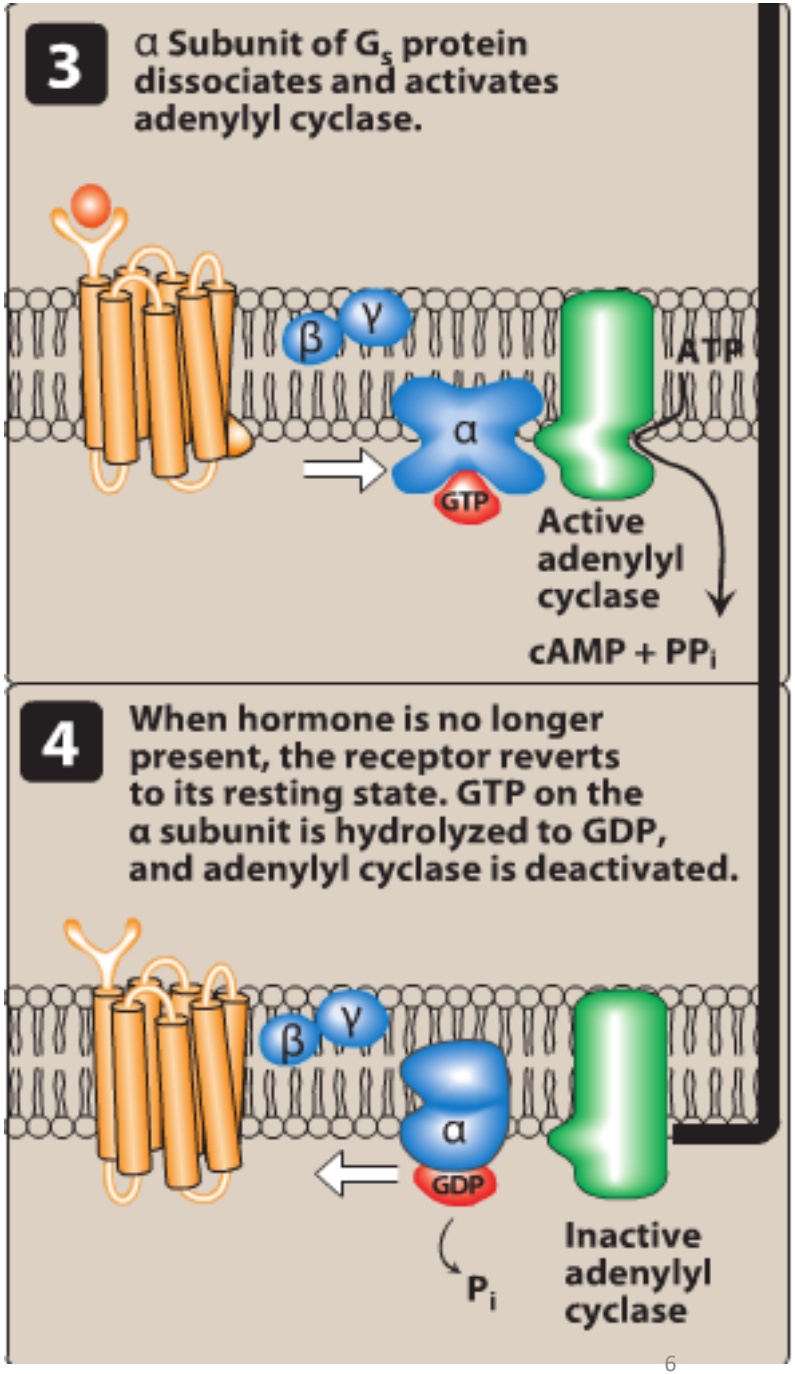


2. G-protein coupled receptors

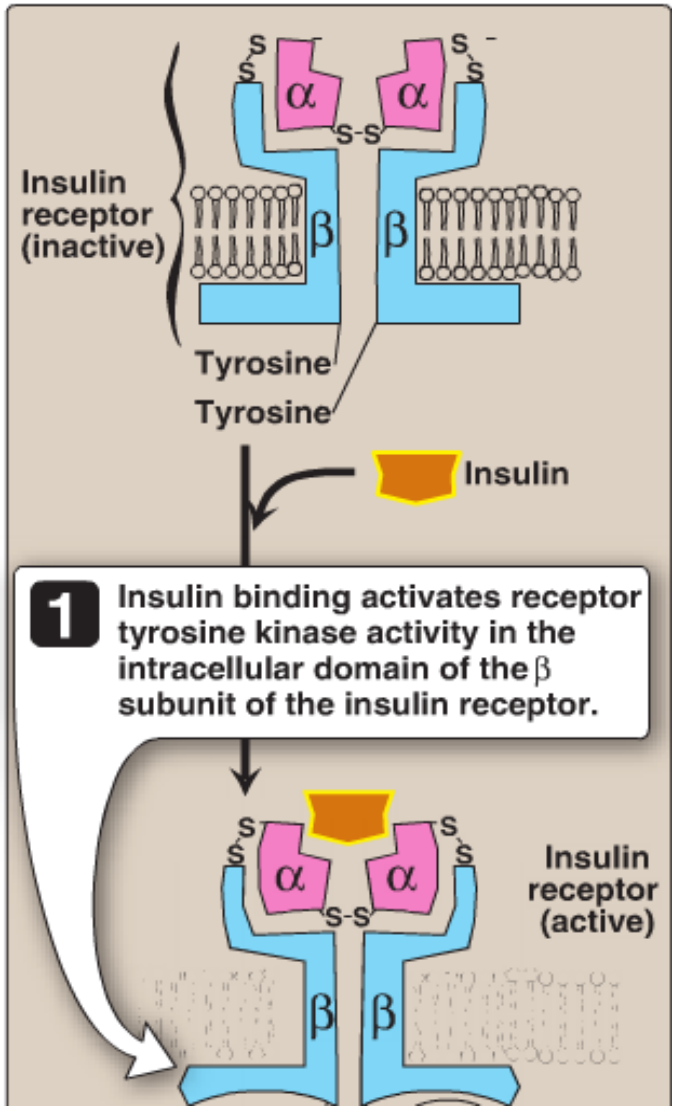


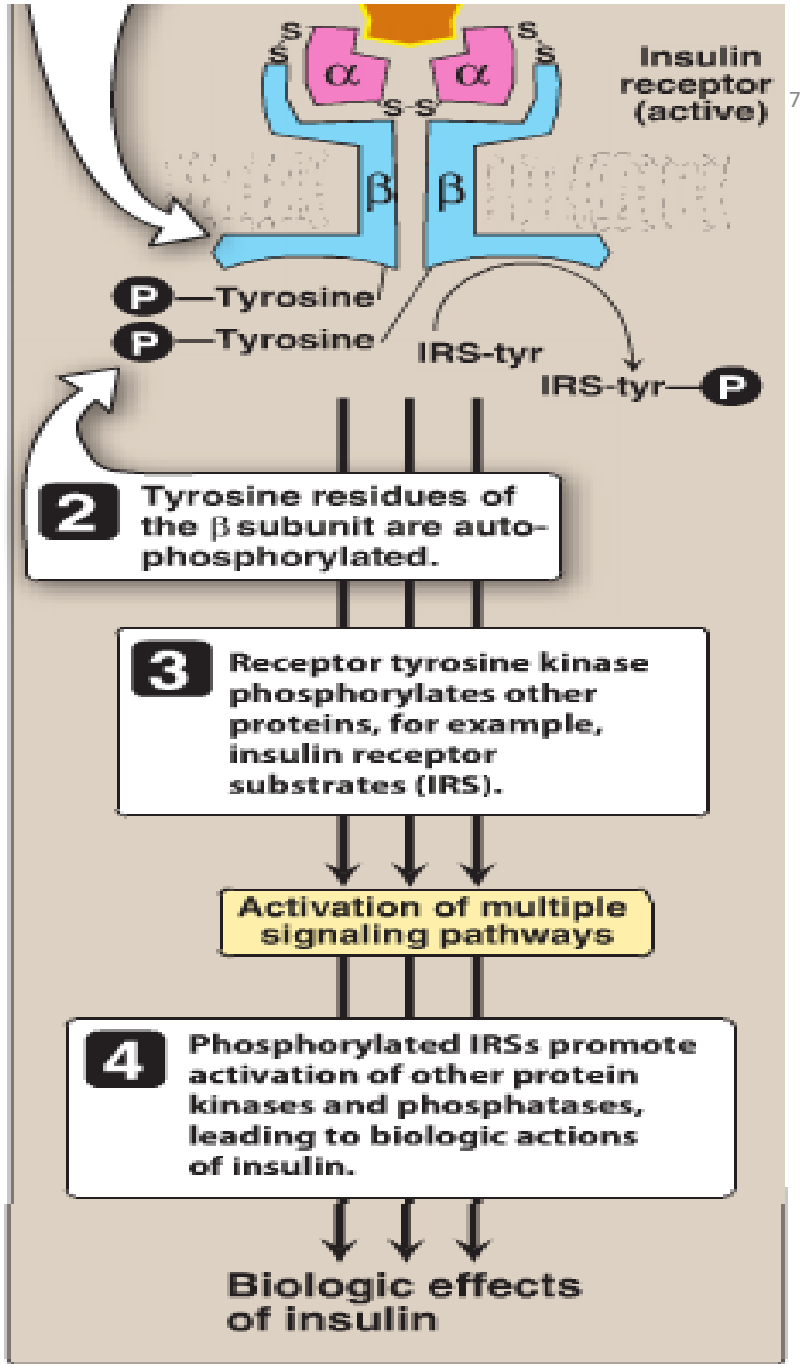


2. G-protein coupled receptors, Cont.



3. Enzyme-linked receptors:





4. Intracellular Receptors

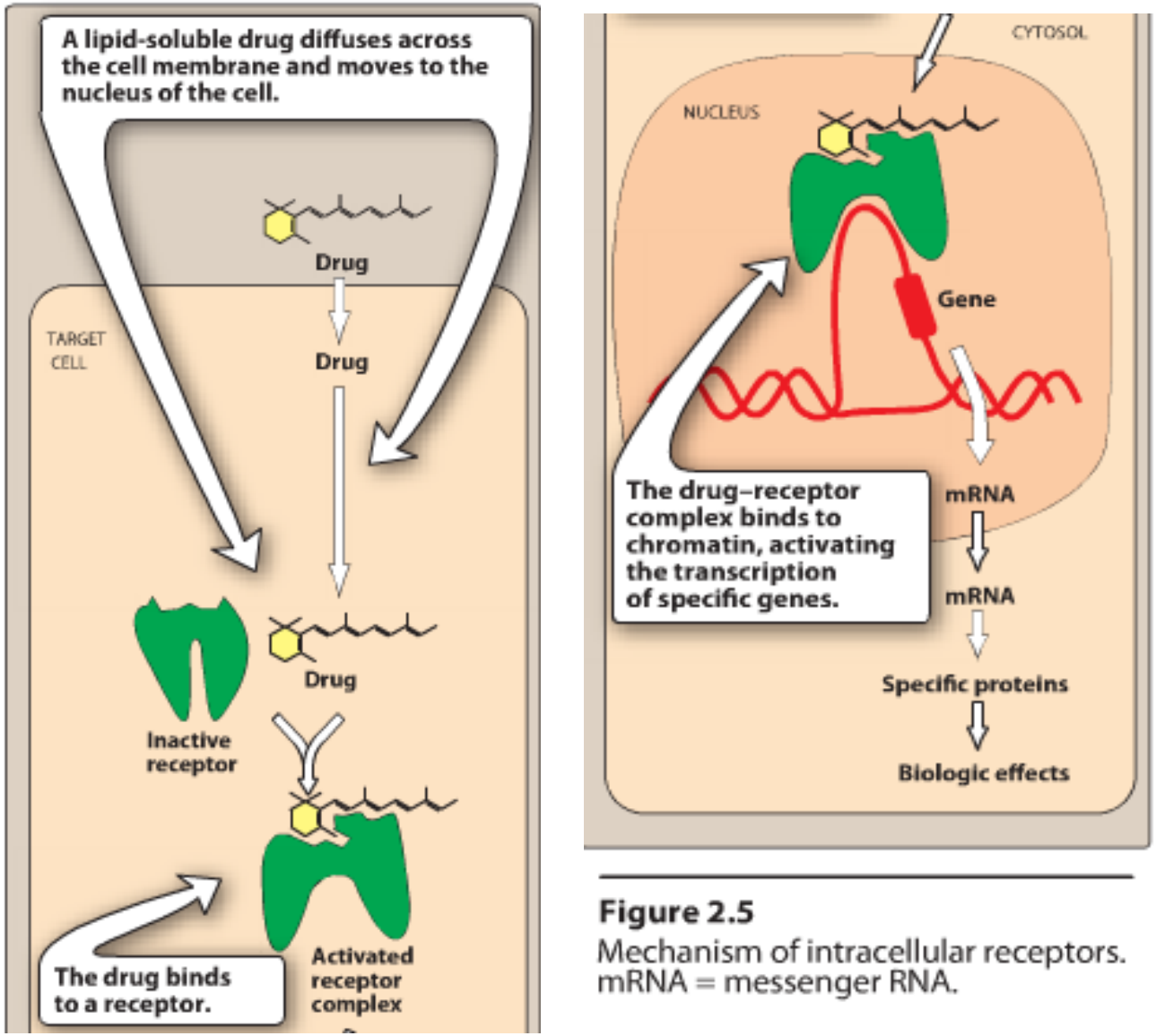


Figure 2.5
Mechanism of intracellular receptors. mRNA = messenger RNA.

III. Properties of drugs

- **Affinity: the chemical forces that cause the drug to associate with the receptor.**

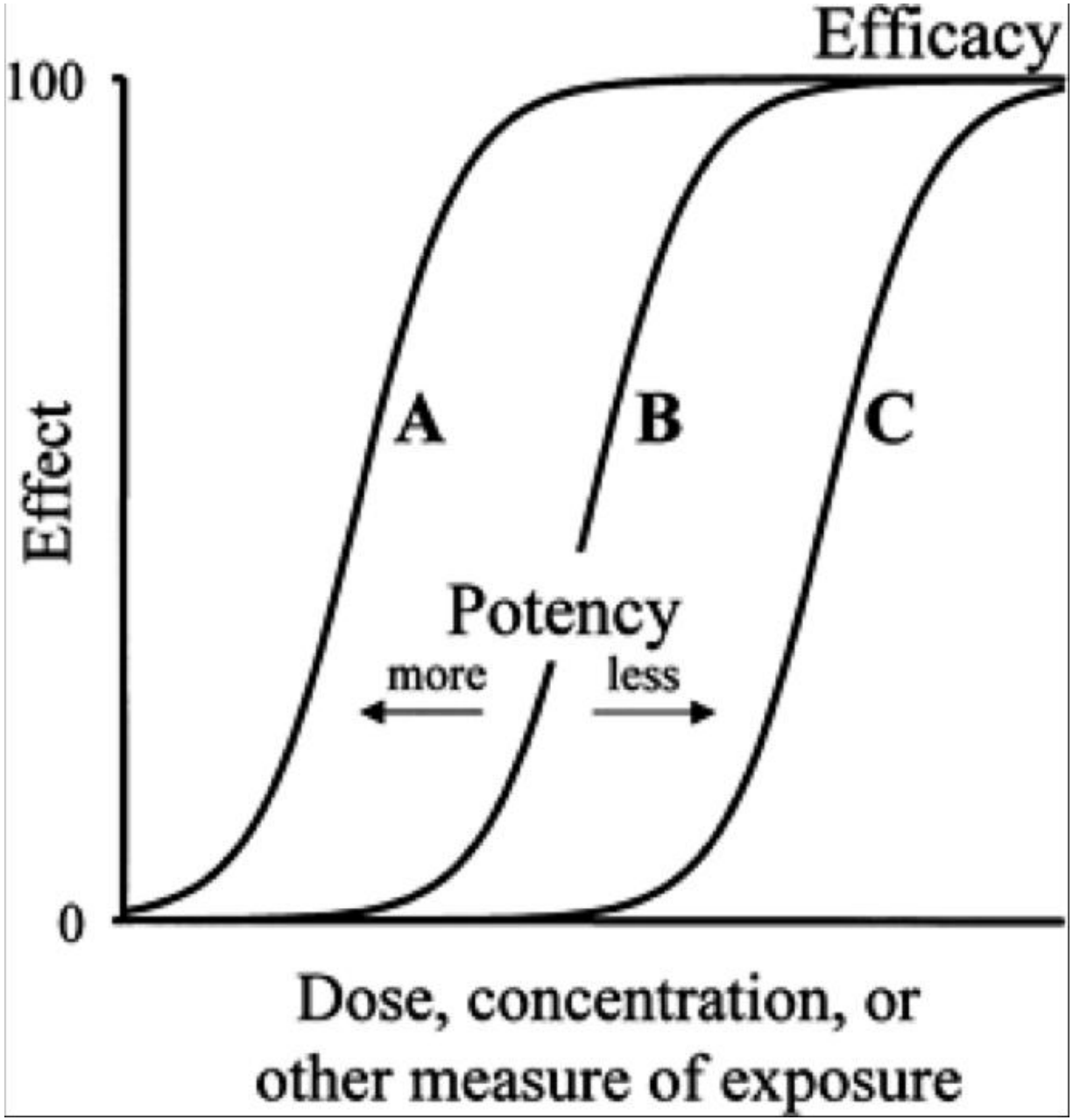


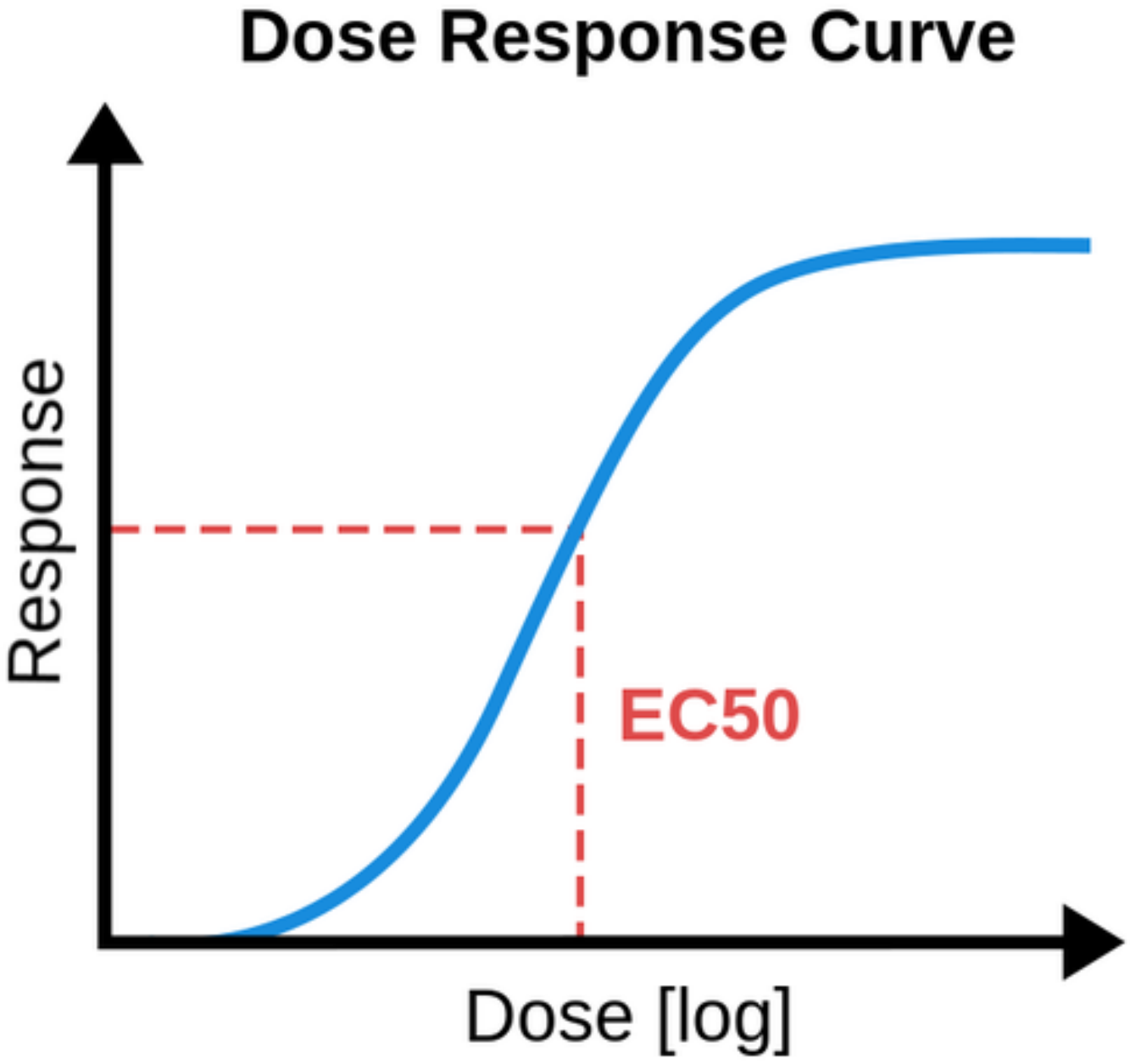
- **Efficacy: the ability of a drug to elicit a response when it interacts with a receptor.**

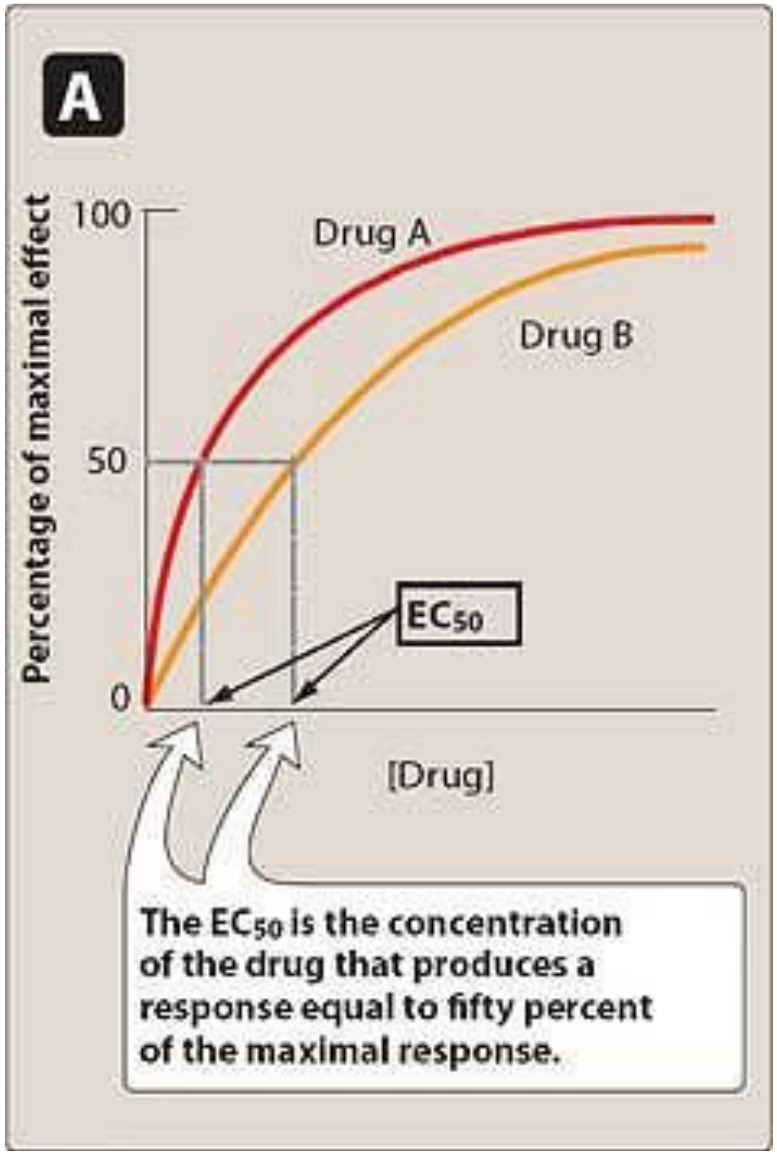


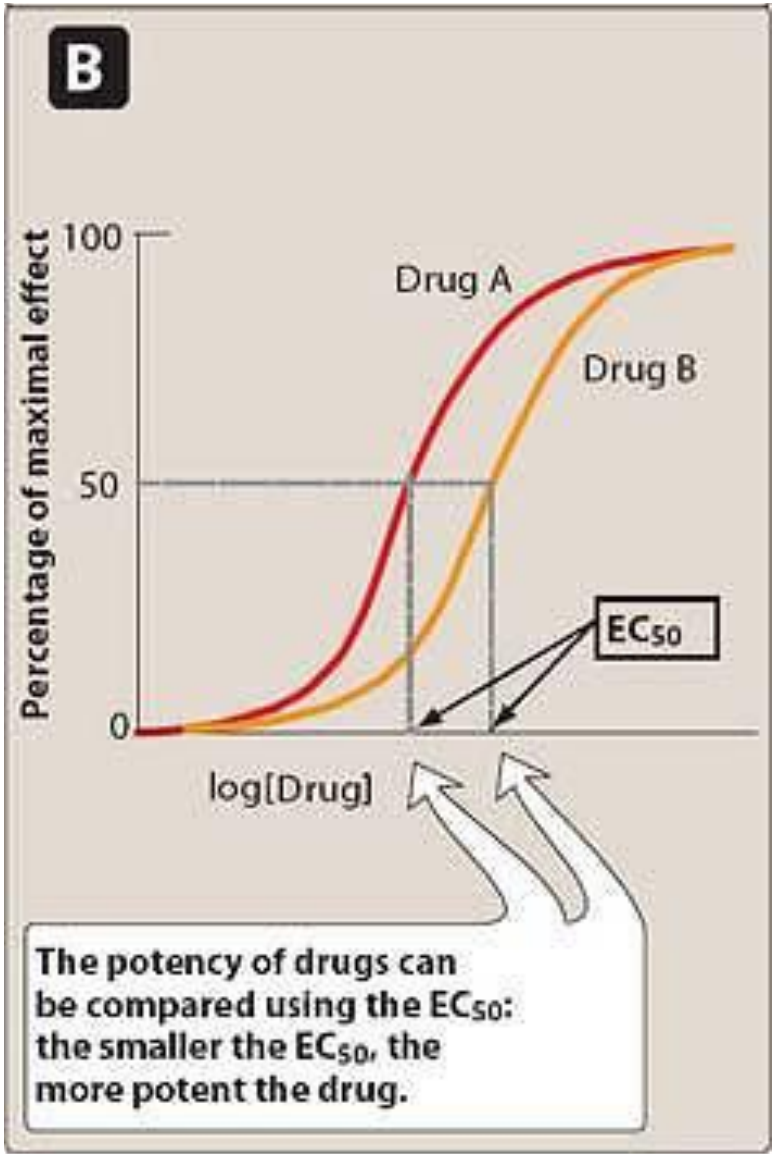
Graded dose–response Relations

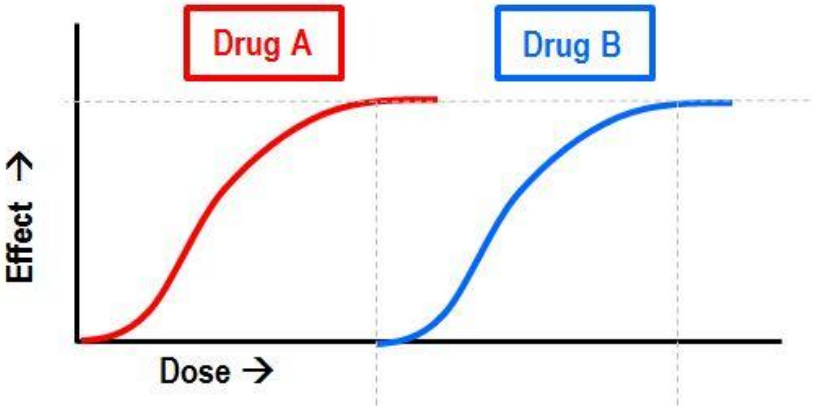
- **Potency:** when the amount of drug necessary to produce an effect of a given magnitude.
  - Dependent upon receptor density, efficiency of the stimulus-response mechanism, affinity and efficacy.
- **Efficacy (Magnitude of effect):** maximal response.
  - Solely dependent upon intrinsic efficacy.

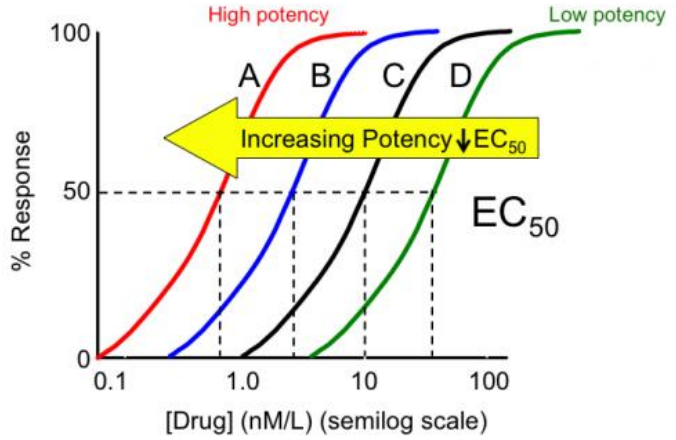

Efficacy
100
Effect
A
B
C
Potency
more
less
0
Dose, concentration, or
other measure of exposure

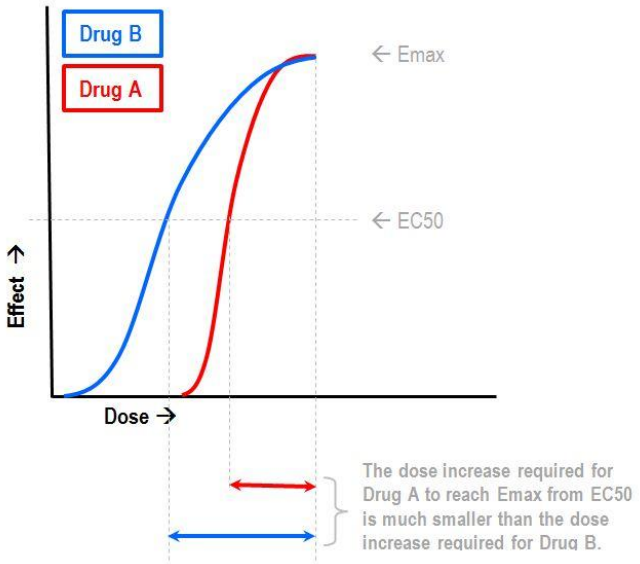

Dose Response Curve
Response
EC50
Dose [log]

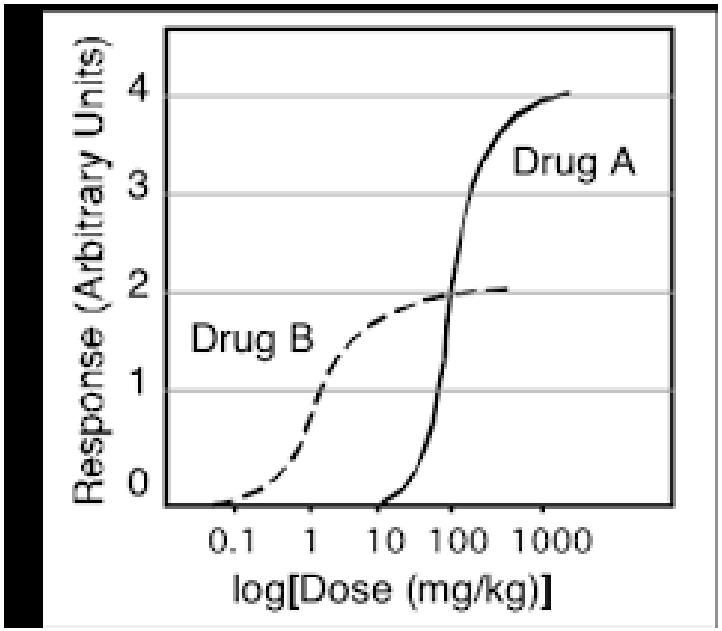

A
Percentage of maximal effect
100
50
0
Drug A
Drug B
EC50
[Drug]
The EC50 is the concentration of the drug that produces a response equal to fifty percent of the maximal response.

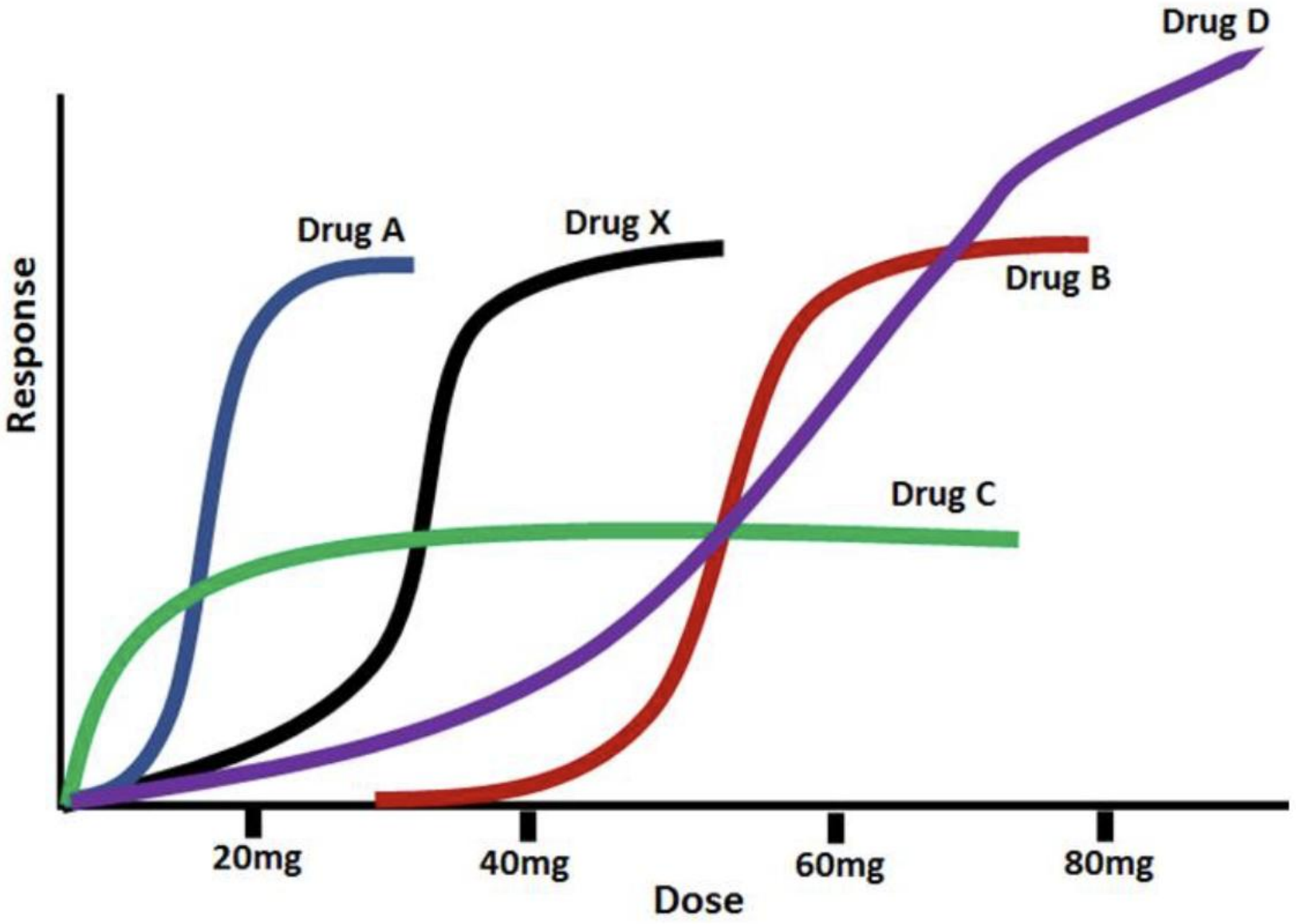

B
Percentage of maximal effect
100
50
0
Drug A
Drug B
EC50
log[Drug]
The potency of drugs can be compared using the EC50: the smaller the EC50, the more potent the drug.


Drug A
Drug B
Effect →
Dose →

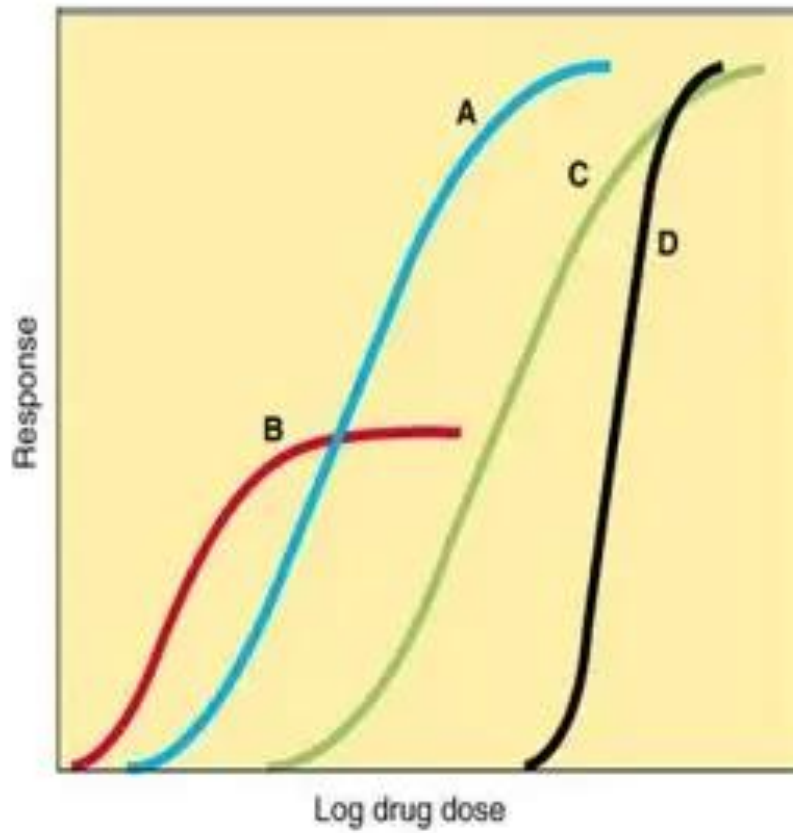

High potency
Low potency
100
50
0
A
B
C
D
Increasing Potency ↓EC50
EC50
% Response
0.1
1.0
10
100
[Drug] (nM/L) (semilog scale)


Drug B
Drug A
← Emax
← EC50
Effect →
Dose →
The dose increase required for Drug A to reach Emax from EC50 is much smaller than the dose increase required for Drug B.


Response (Arbitrary Units)
4
3
2
1
0
Drug A
Drug B
0.1
1
10
100
1000
log[Dose (mg/kg)]


Drug D
Drug A
Drug X
Drug B
Drug C
Response
20mg
40mg
60mg
80mg
Dose

Which drug is the most potent? *

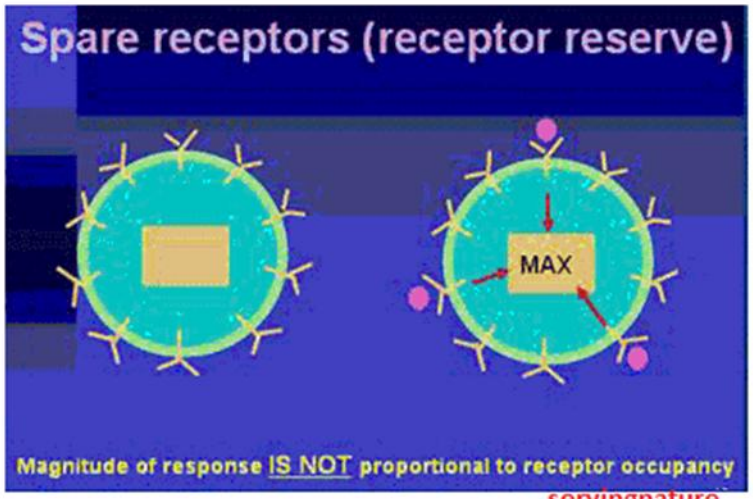


- ○ A
- ○ B
- ○ C
- ○ D
- ○ All with the same potency

- A drug with greater efficacy is more therapeutically beneficial than one that is more potent.

- Maximal efficacy of a drug assumes that all receptors are occupied by the drug, and no increase in response will be observed if more drugs are added.

- This concept holds true only if there are no "spare receptors" present.





IV. Concepts to remember!

- **Threshold:** Dose that produces a just-noticeable effect.

- **ED_{50}:** Dose that produces a 50% of maximum response.

- **EC_{50}**: refers to the concentration of a drug which induces a response halfway between the baseline and maximum after a specified exposure time.

- **Ceiling:** Lowest dose that produces a maximal effect.



Dose-response curve

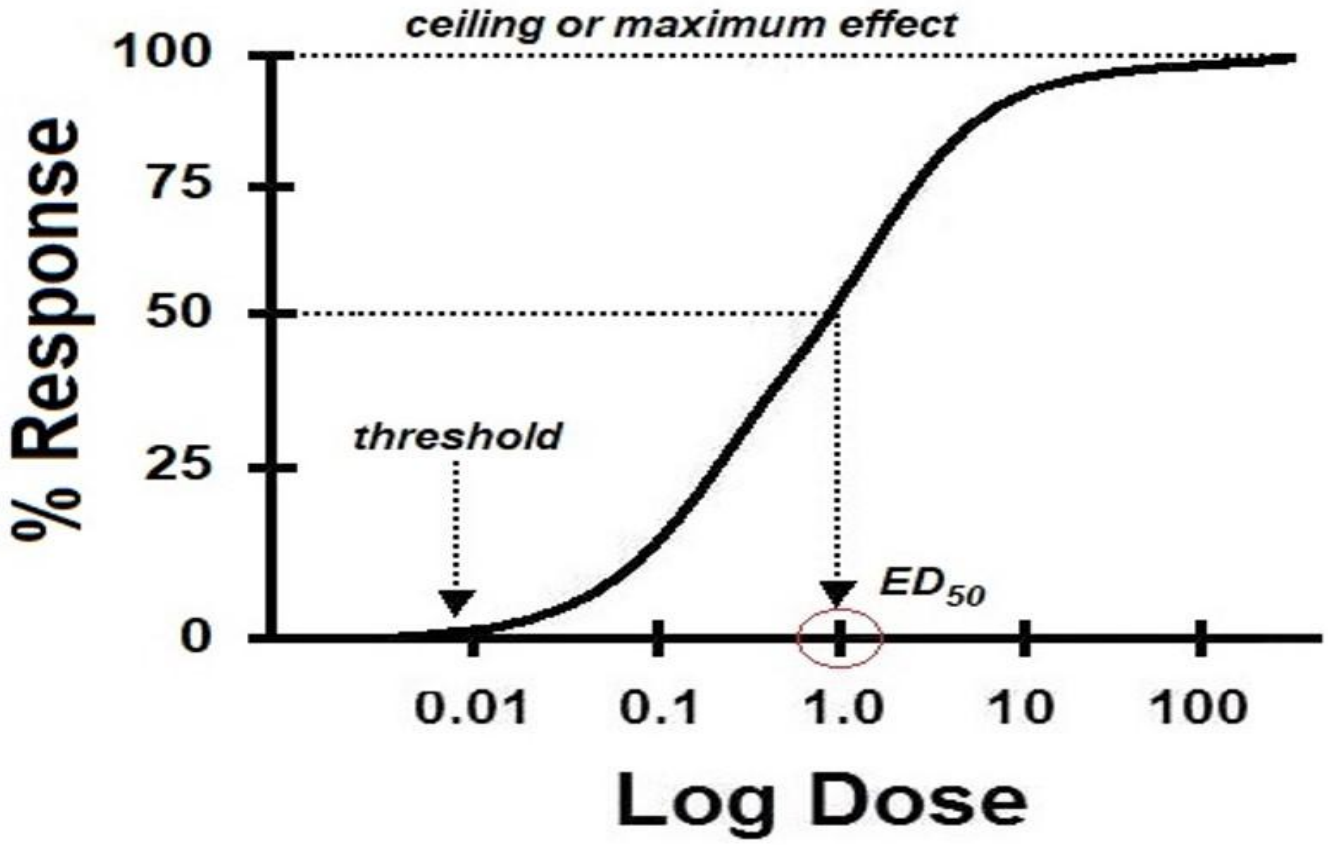
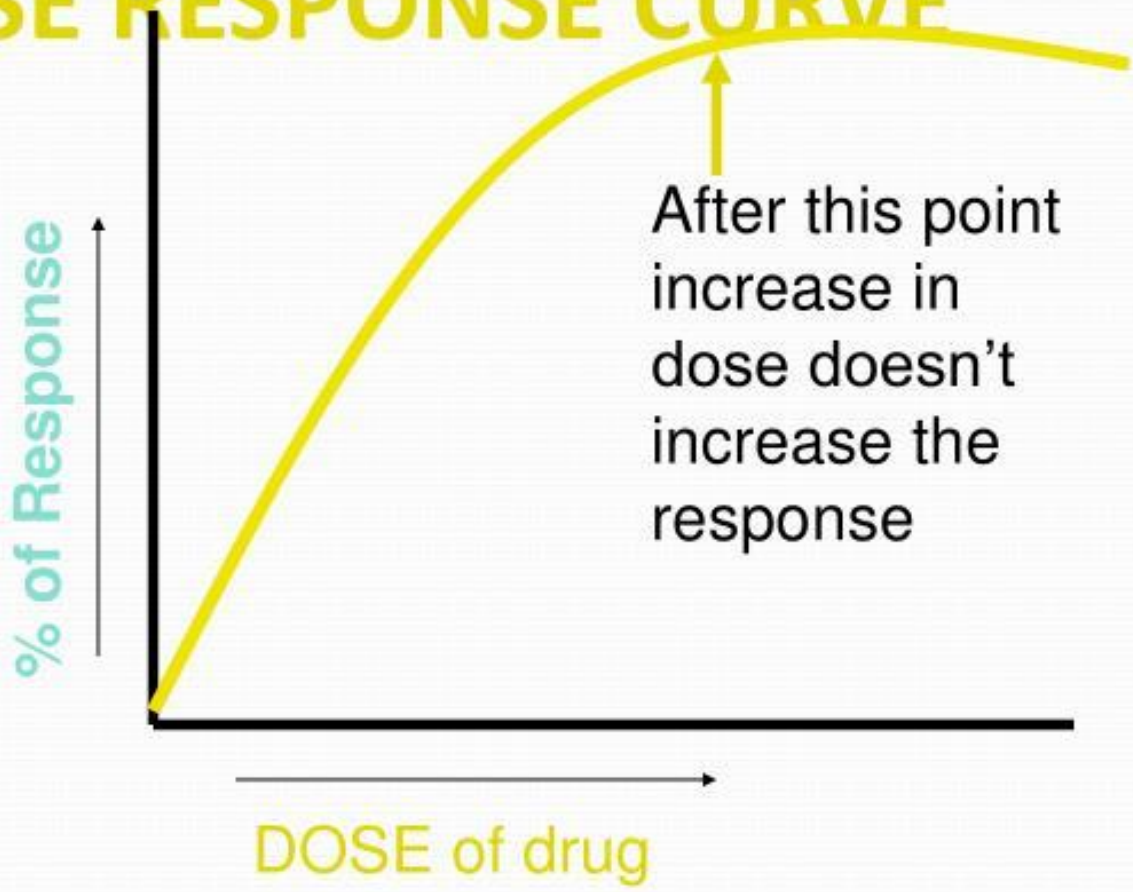
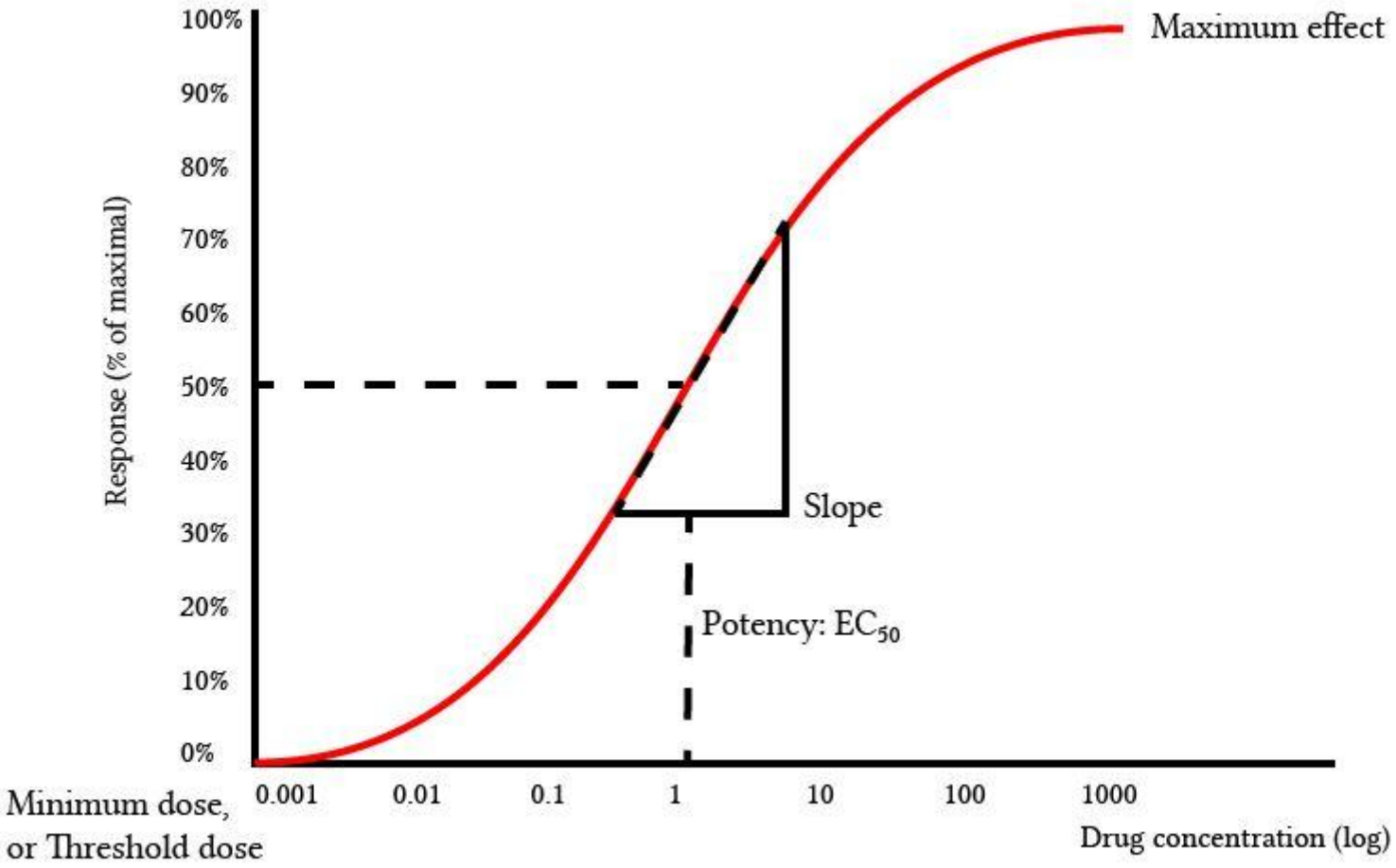


Figure 6.8. Threshold and Ceiling Doses

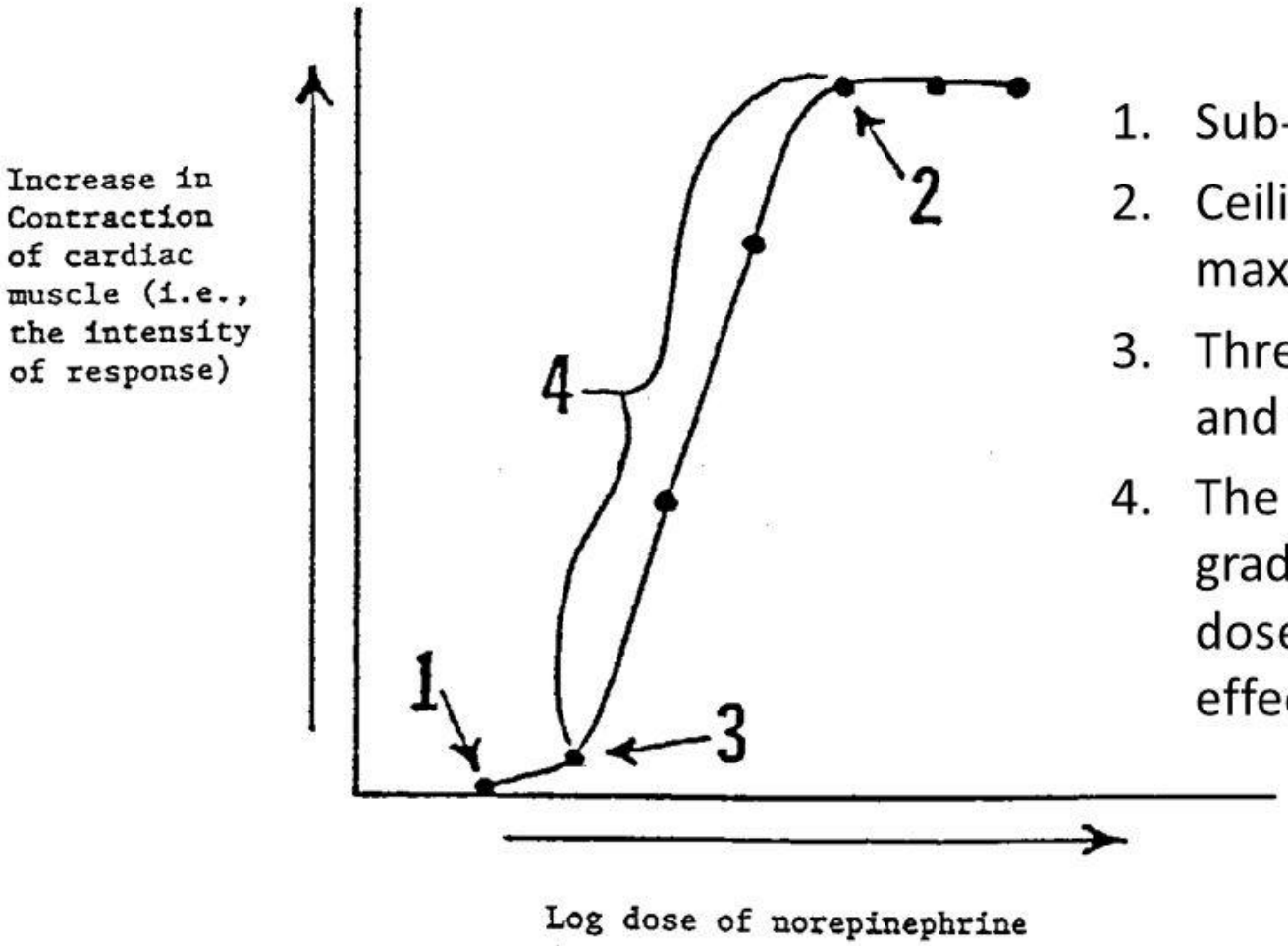
Original drawing by Raymond M. Quock

DOSE RESPONSE CURVE




100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Response (% of maximal)
Maximum effect
Slope
Potency: EC_{50}
Minimum dose,
or Threshold dose
0.001
0.01
0.1
1
10
100
1000
Drug concentration (log)

Dose-Response Curve



1. Sub-effective dose
2. Ceiling effect, or maximal effect
3. Threshold effect and threshold dose
4. The plot shows a graded response - dose-dependent effect

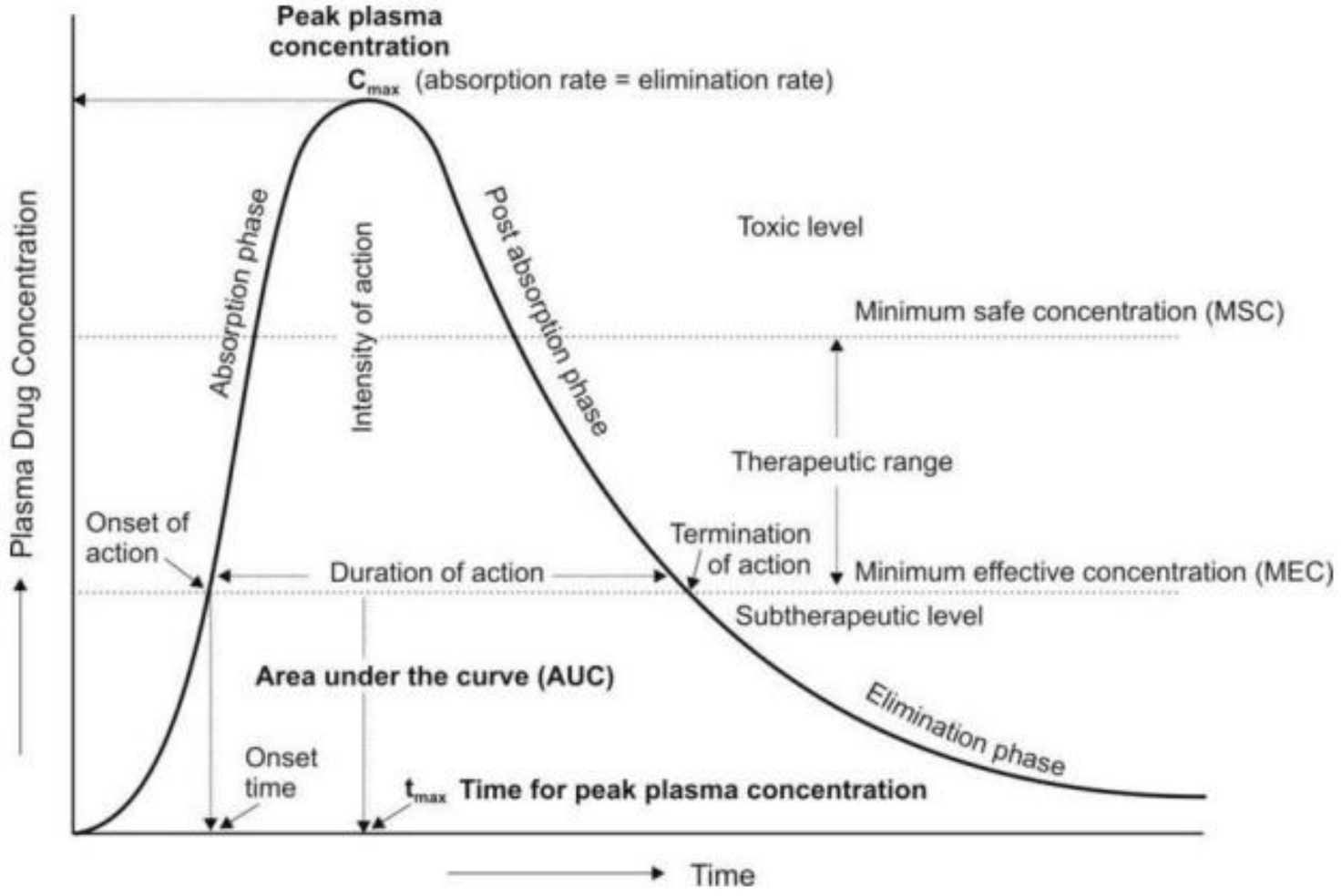
Important parameters:

- **Onset**: The time it takes for the drug to elicit a therapeutic response.

- **Peak**: The time it takes for a drug to reach its maximum therapeutic response.

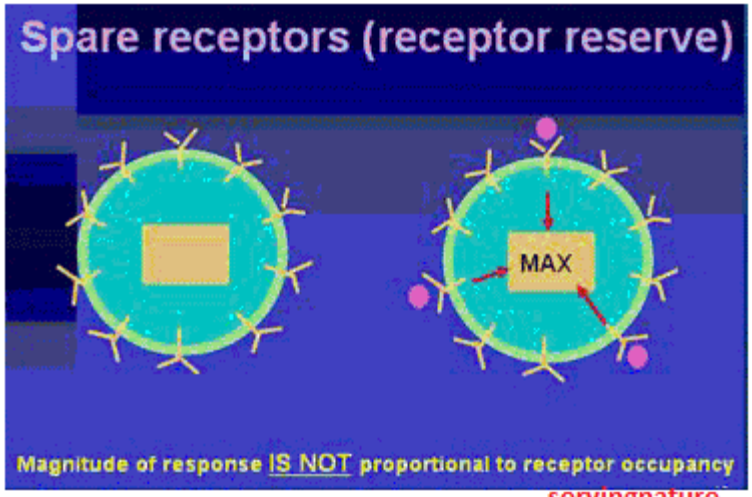
- **Duration**: The time a drug concentration is sufficient to elicit a therapeutic response

Pharmacodynamic Parameters



Spare receptors

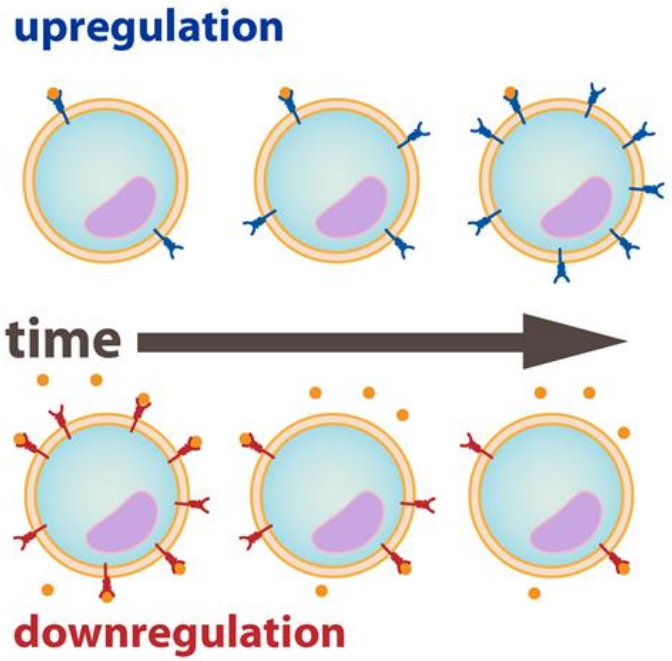
- only a fraction of the total receptors for a specific ligand may need to be occupied to elicit a maximal response from a cell.
- Systems that exhibit this behavior are said to have spare receptors.
- Spare receptors are exhibited by:
- insulin receptors: 99 percent of the receptors are “spare.”
- β-adrenoceptors in the heart: 5 to 10 percent of the total β-adrenoceptors are spare.



Desensitization and down-regulation of receptors:

- Repeated or continuous administration of an agonist (or an antagonist) may lead to changes in the responsiveness of the receptor.
- When repeated administration of a drug results in a diminished effect, the phenomenon is called **tachyphylaxis**.
- The receptor becomes desensitized to the action of the drug.
- In this phenomenon, the receptors are still present on the cell surface but are unresponsive to the ligand.

- Receptors can also be down-regulated in the presence of continual stimulation.
- Binding of the agonist results in molecular changes in the membrane-bound receptors, such that the receptor undergoes endocytosis and is sequestered within the cell, unavailable for further agonist interaction.



Tolerance vs Tachyphylaxis

- **Tolerance**: Tolerance is the gradual adaptation of the body to a drug, resulting in the need for increased doses to achieve the same therapeutic effect that was initially obtained with a lower dose. Example, Morphine tolerance.

- **Tachyphylaxis**: This refers to a rapid decrease in the effectiveness of a drug after its initial administration. It can occur after just a few doses and is characterized by a quick onset of diminished response, often within a short time frame.

V. Classification of a drug based on drug-receptor interactions

1. Agonist: If a drug binds to a receptor and produces a maximal biologic response that mimics the response to the endogenous ligand, it is known as a full agonist.

e.g. phenylephrine is an agonist at α1-adrenoceptors, because it produces effects that resemble the action of the endogenous ligand, norepinephrine.

- In general, a full agonist has a strong affinity for its receptor and good efficacy

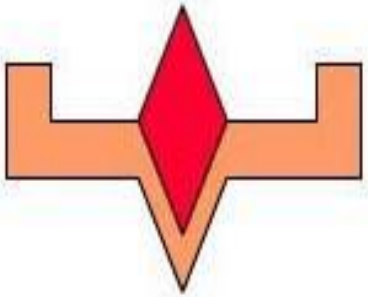
cont.

2. Partial agonists:

- Drug that, no matter how high the dose, cannot produce a full response.
- Partial agonists have efficacies (intrinsic activities) greater than zero but less than that of a full agonist.
- Affinity of a partial agonist may be greater than, less than, or equivalent to that of a full agonist.
- Under appropriate conditions, a partial agonist may act as an antagonist of a full agonist. (i.e. Partial agonists have both agonist and antagonist properties)

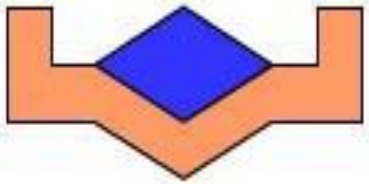
- Example, aripiprazole, an atypical neuroleptic agent, is a partial agonist at selected dopamine receptors.

Full Agonist

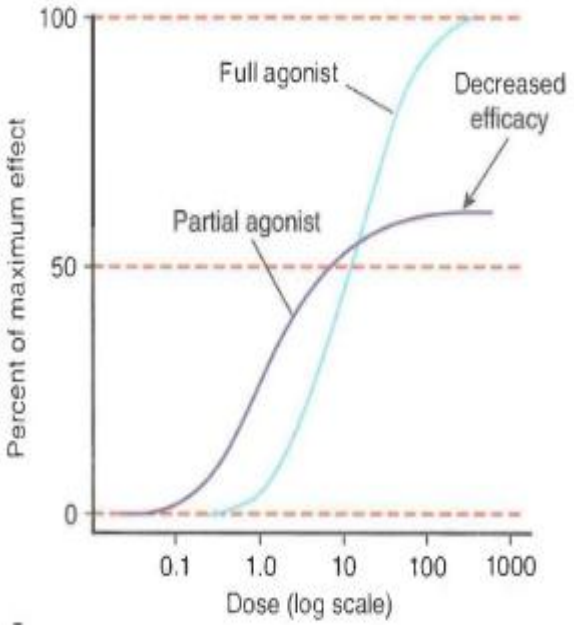


The full agonist can induce a conformational change in the receptor leading to a maximal effect. The ability to induce changes in the receptor conformation leading to activation is a measure of the intrinsic activity.

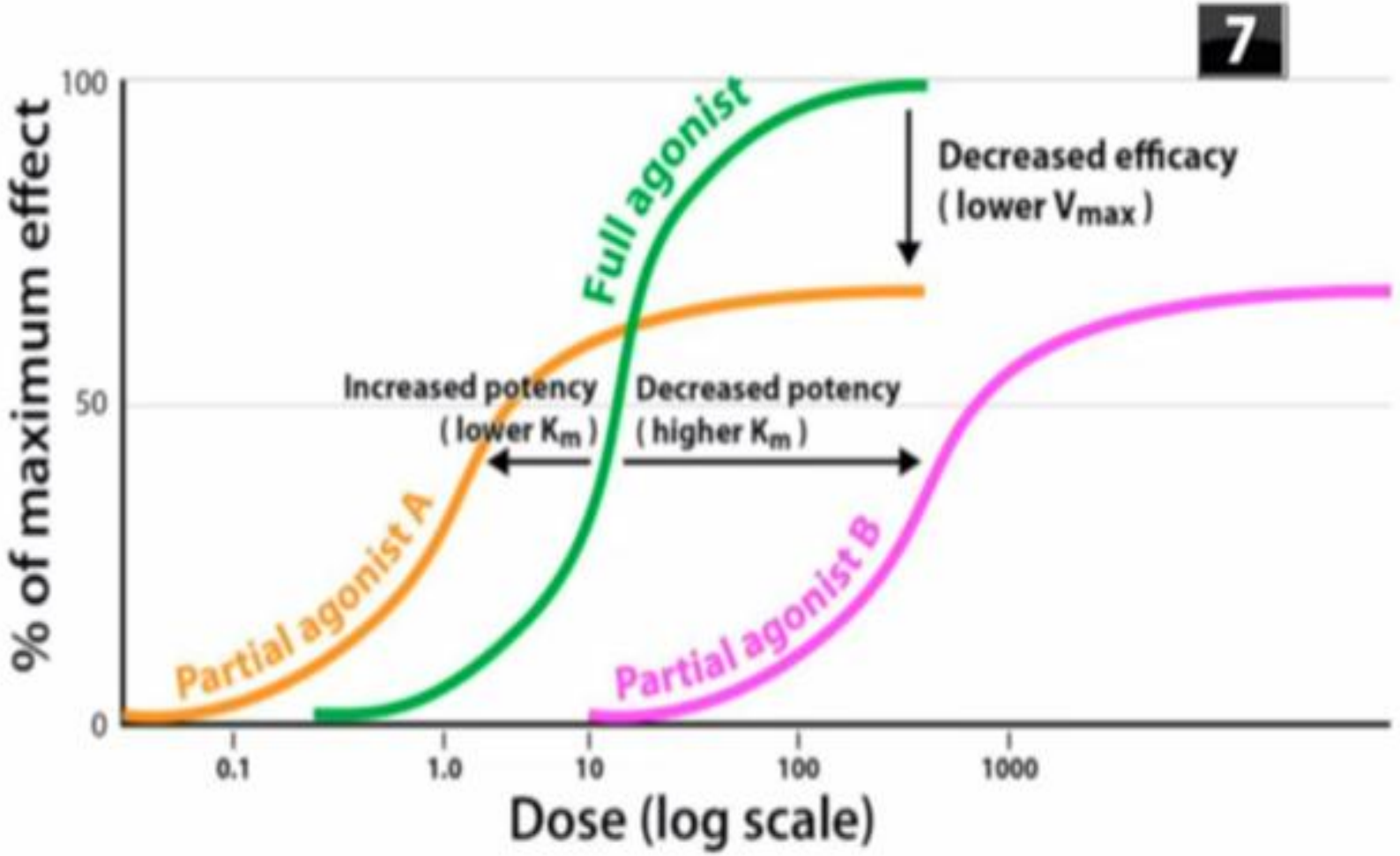
Partial Agonist



Partial agonists can induce some degree of receptor activation but not of sufficient magnitude for a maximal response

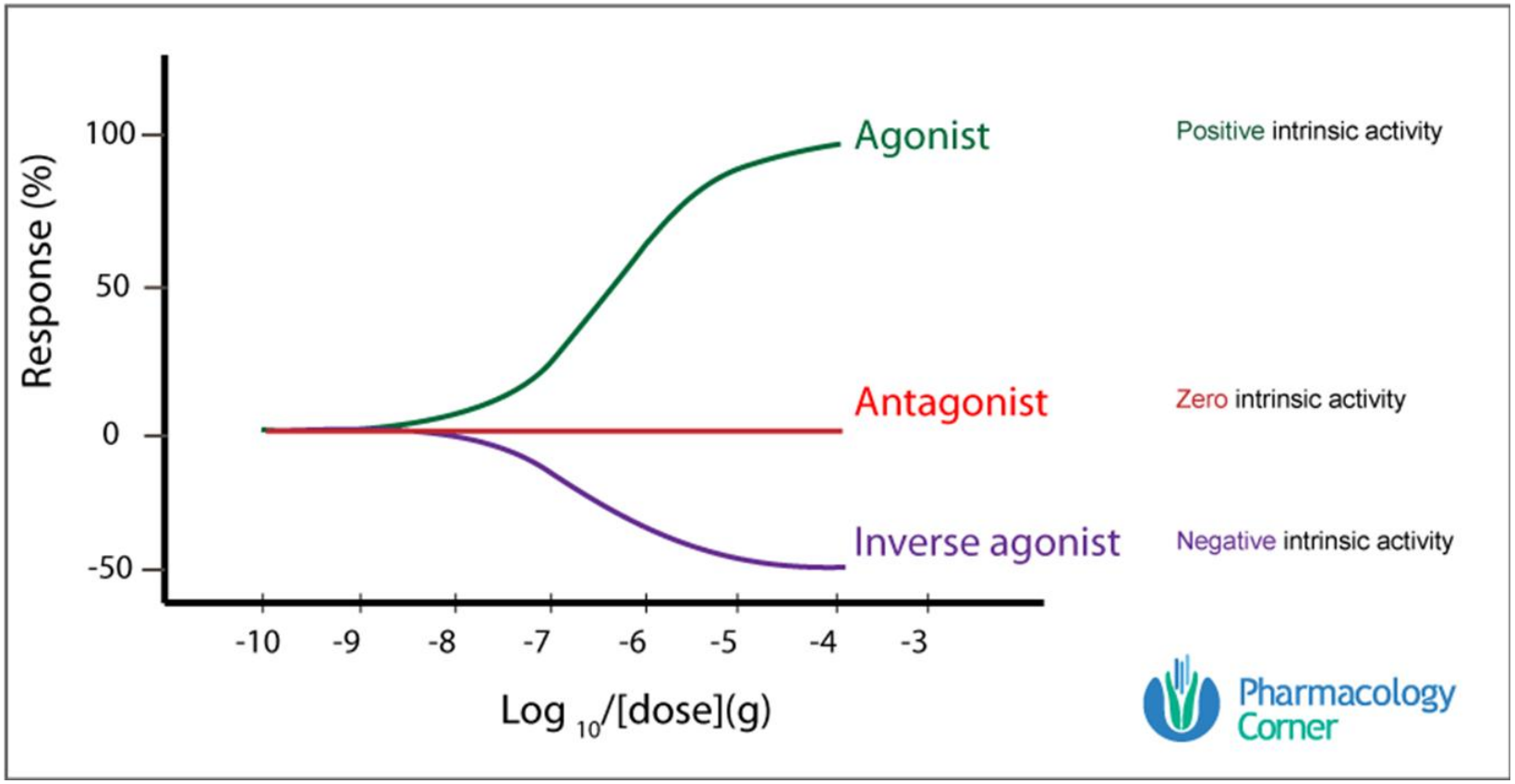


Partial Agonist & full Agonist



3. Inverse agonist:

- Drug that binds to a receptor to produce an effect opposite that of an agonist.
- Stabilizes receptors in the inactive state.



Partial, Full & Inverse Agonists

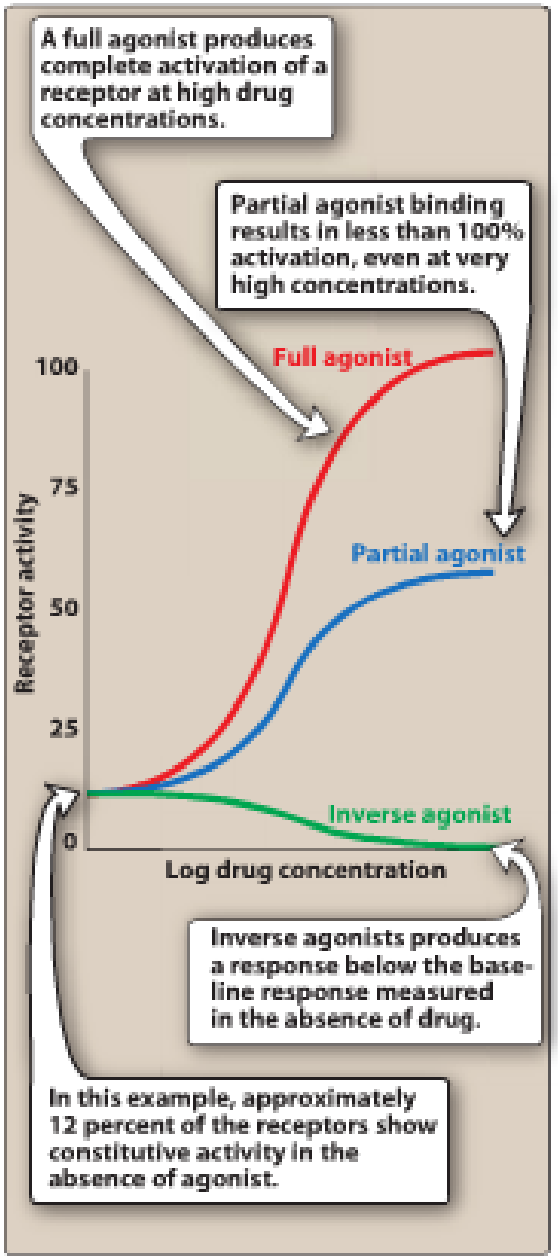


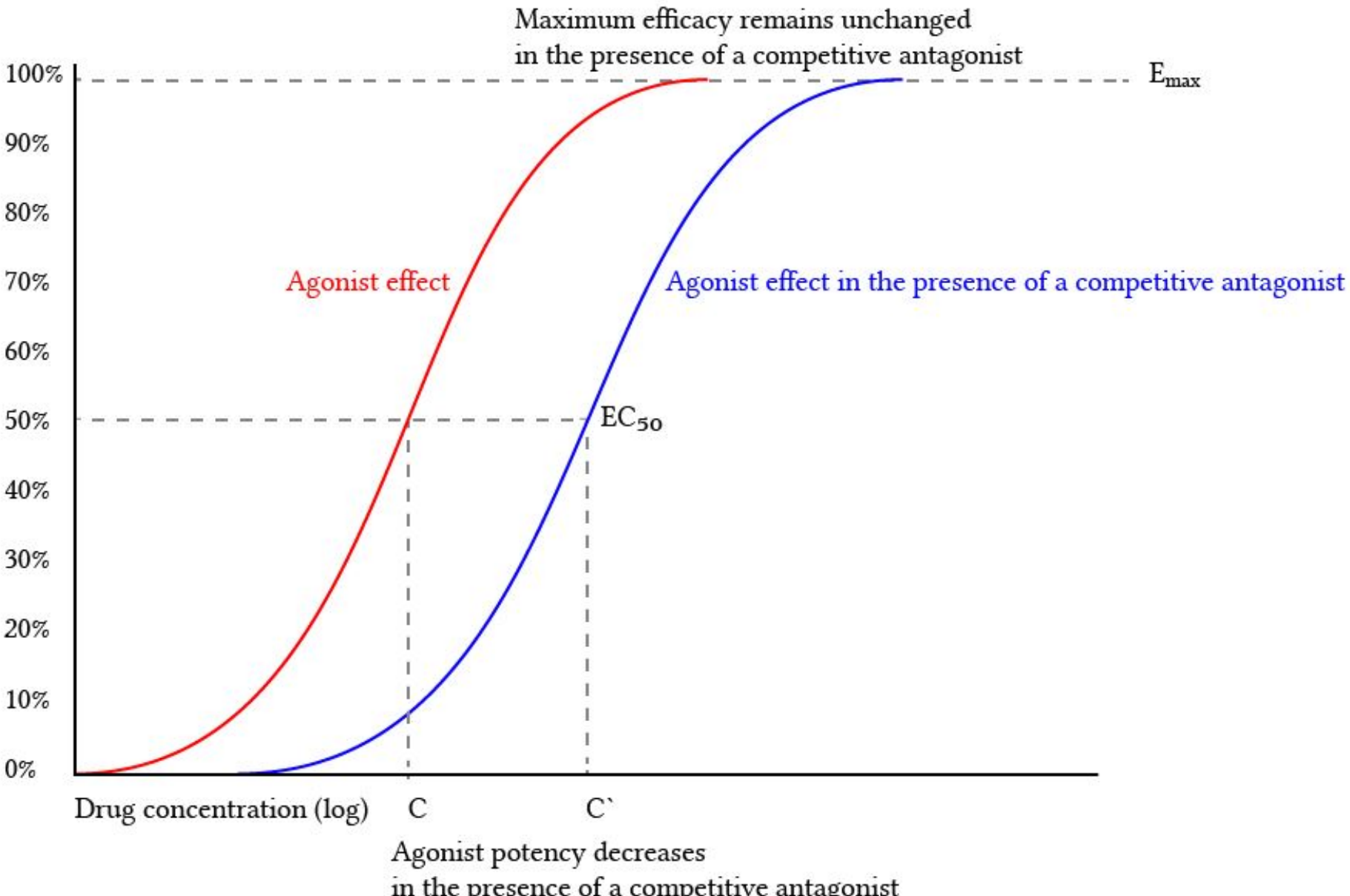
Figure 2.10
Effects of full agonists, partial agonists, and inverse agonist on receptor activity.

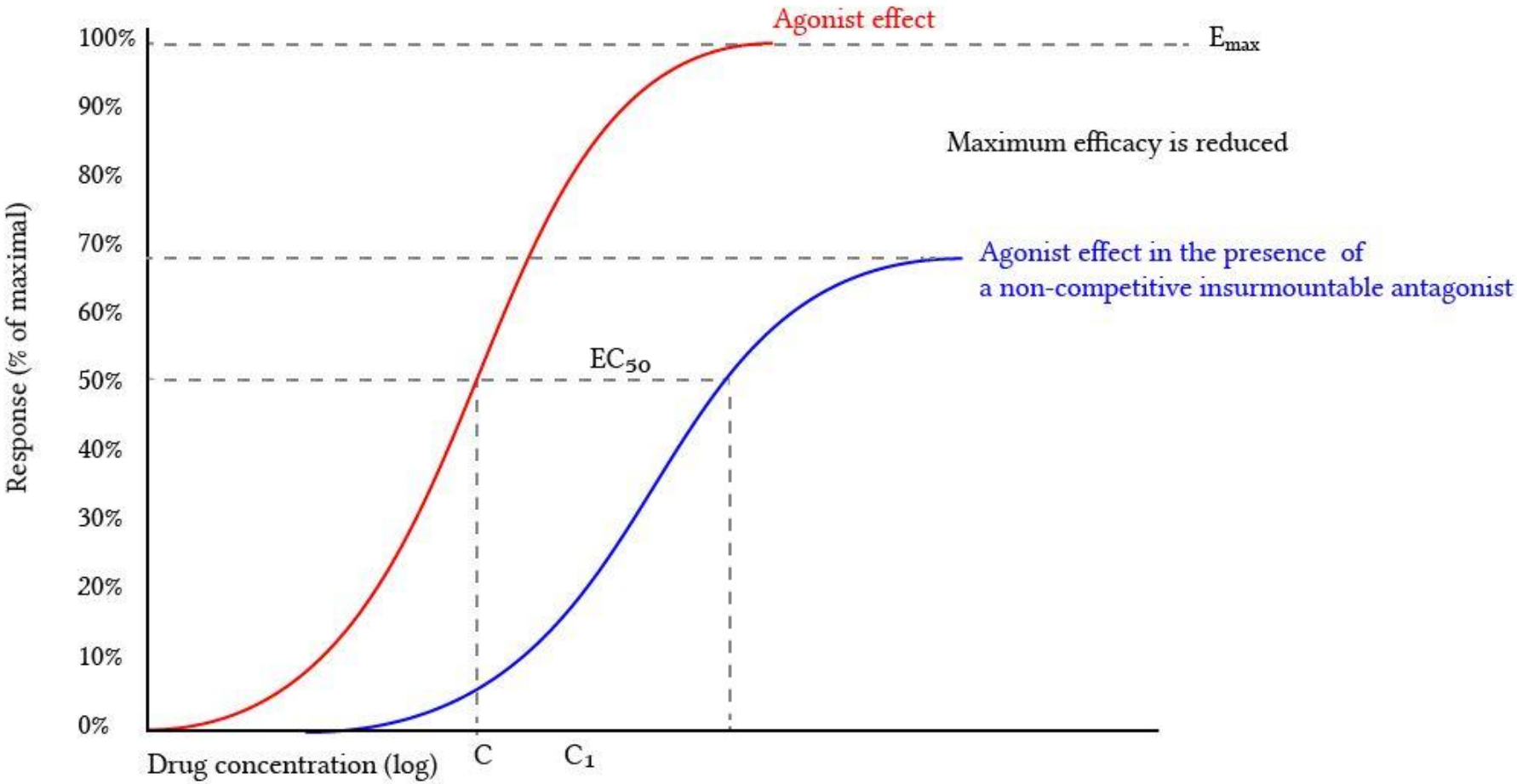
4. Antagonists

- **Antagonist:** drug that binds to receptors but cannot initiate a cellular response, but prevent agonists from producing a response; affinity, but no efficacy. Antagonists maintain the active-inactive equilibrium. Antagonists might be competitive or non-competitive;

1. **Competitive Antagonists:** Antagonist binds to same site as agonist in a reversible manner.
2. **Noncompetitive Antagonists:** Antagonist binds to the same site as agonist irreversibly.

A ***dextral shift*** (rightward shift) of an agonist concentration-response curve occurs when an antagonist decreases the apparent potency of an agonist, meaning a higher concentration of the agonist is required to produce the same biological effect.

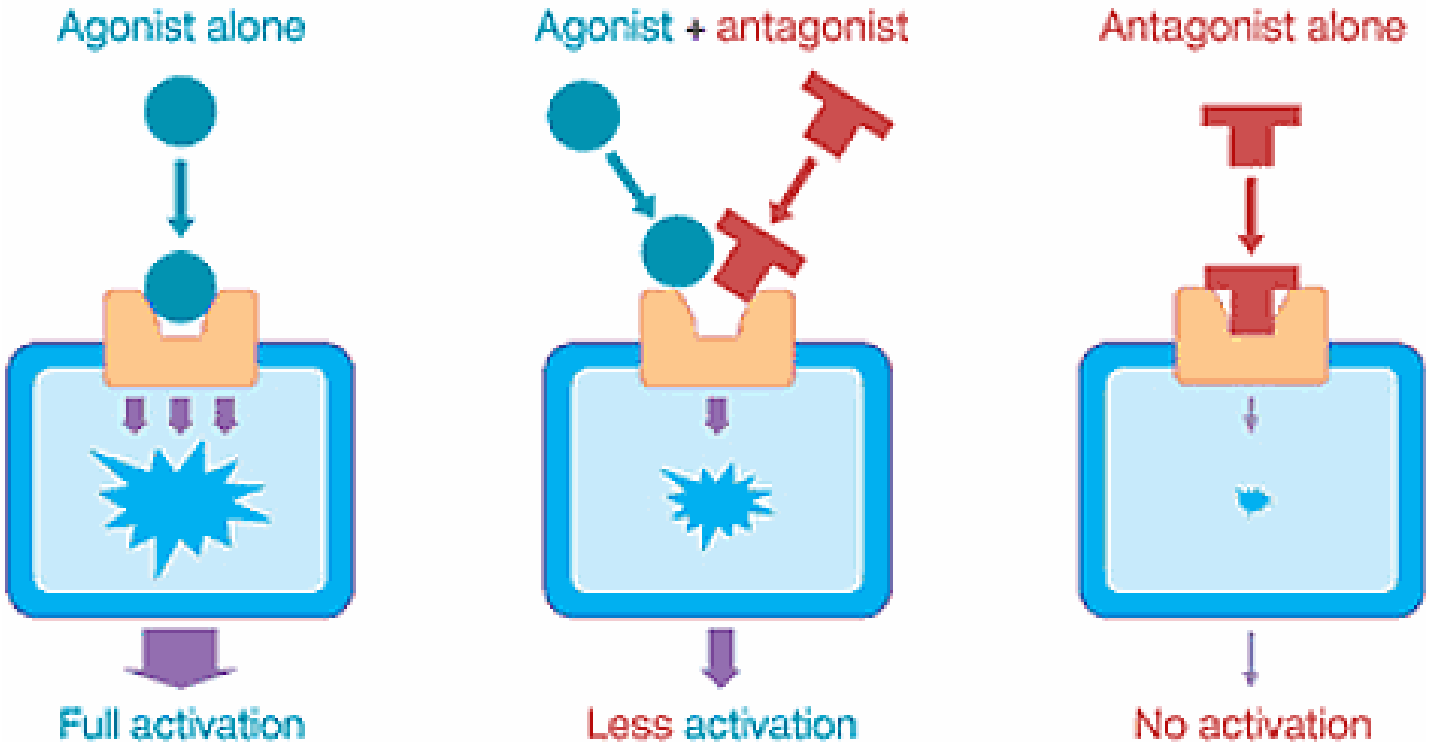



Agonist effect
100%
E_{max}
90%
Maximum efficacy is reduced
80%
70%
Agonist effect in the presence of
a non-competitive insurmountable antagonist
60%
EC_{50}
50%
40%
30%
20%
10%
0%
Response (% of maximal)
Drug concentration (log)
C
C_1

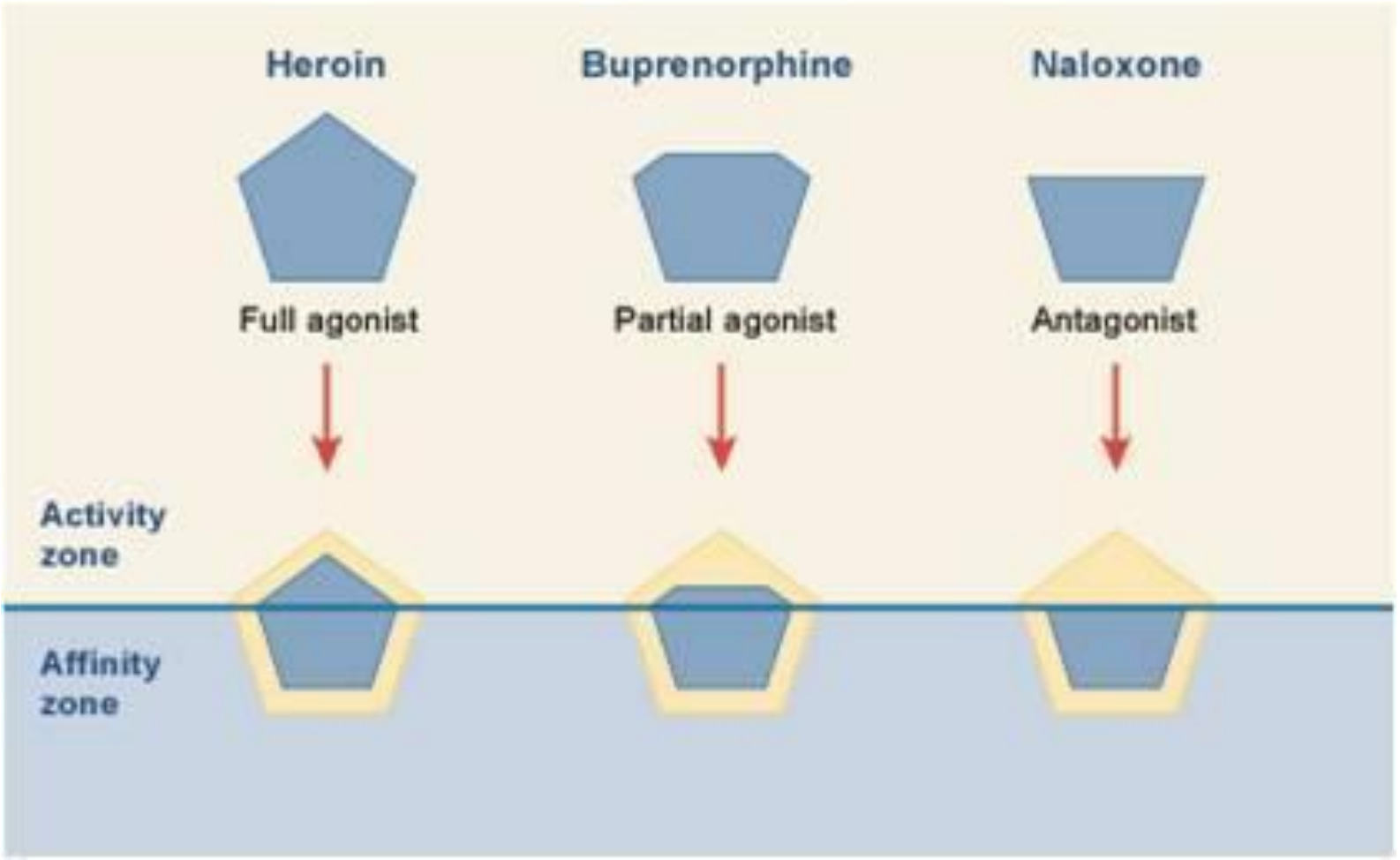
Agonists and Antagonists

Agonists Drugs that occupy receptors and activate them.

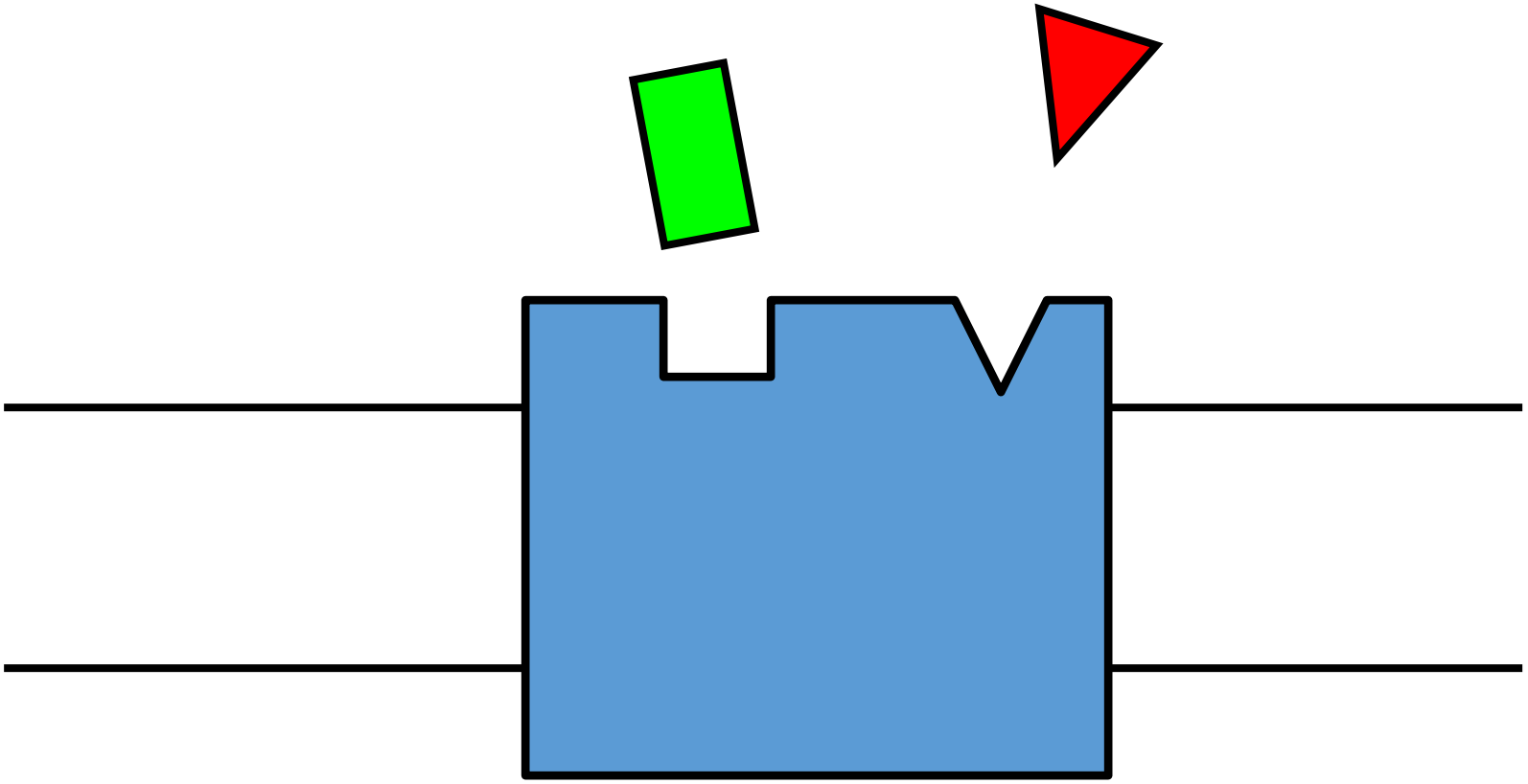
Antagonists Drugs that occupy receptors but do not activate them. Antagonists block receptor activation by agonists.



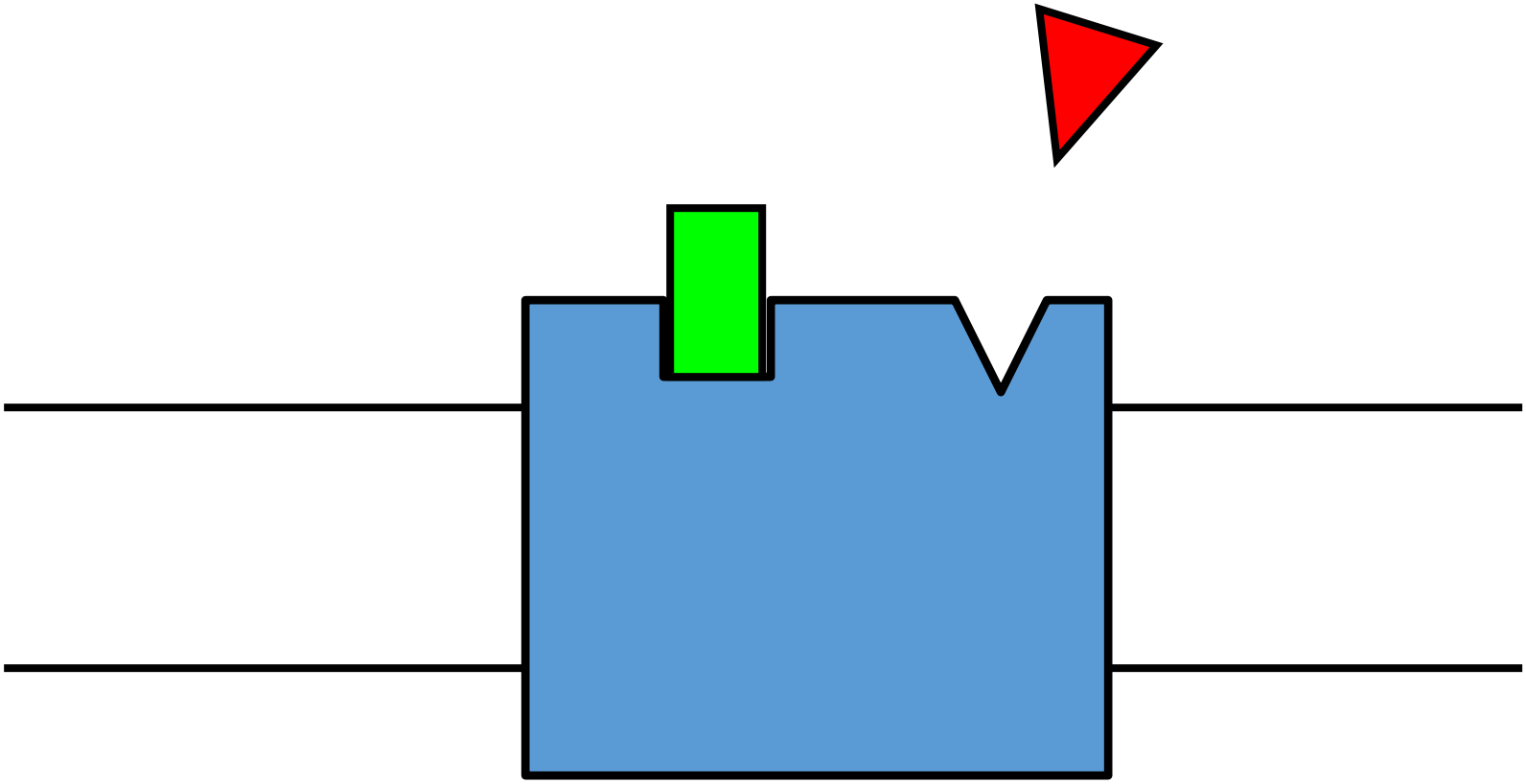
Agonists & Antagonists



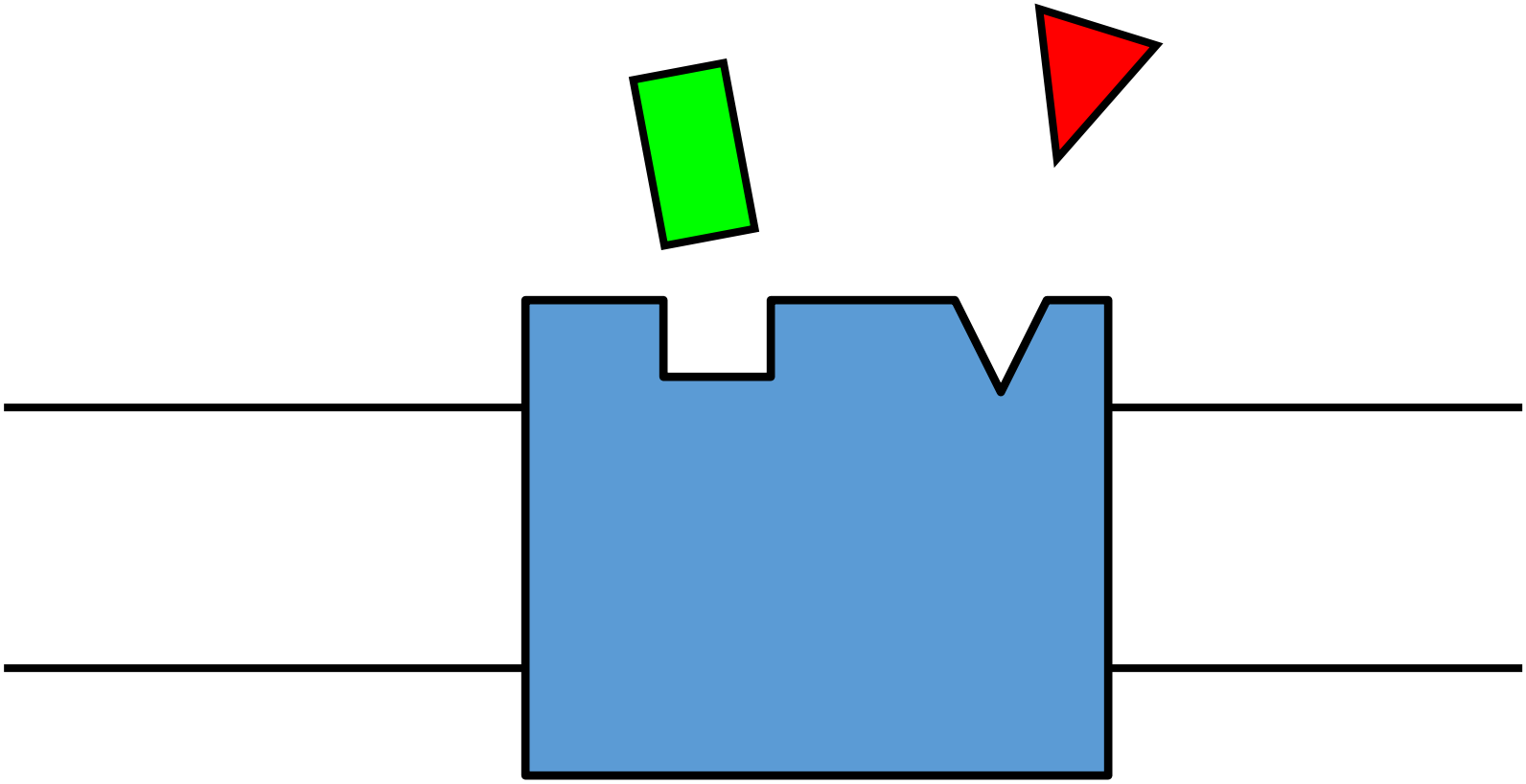
- **Allosteric:** Antagonist and agonist bind to different site on same receptor



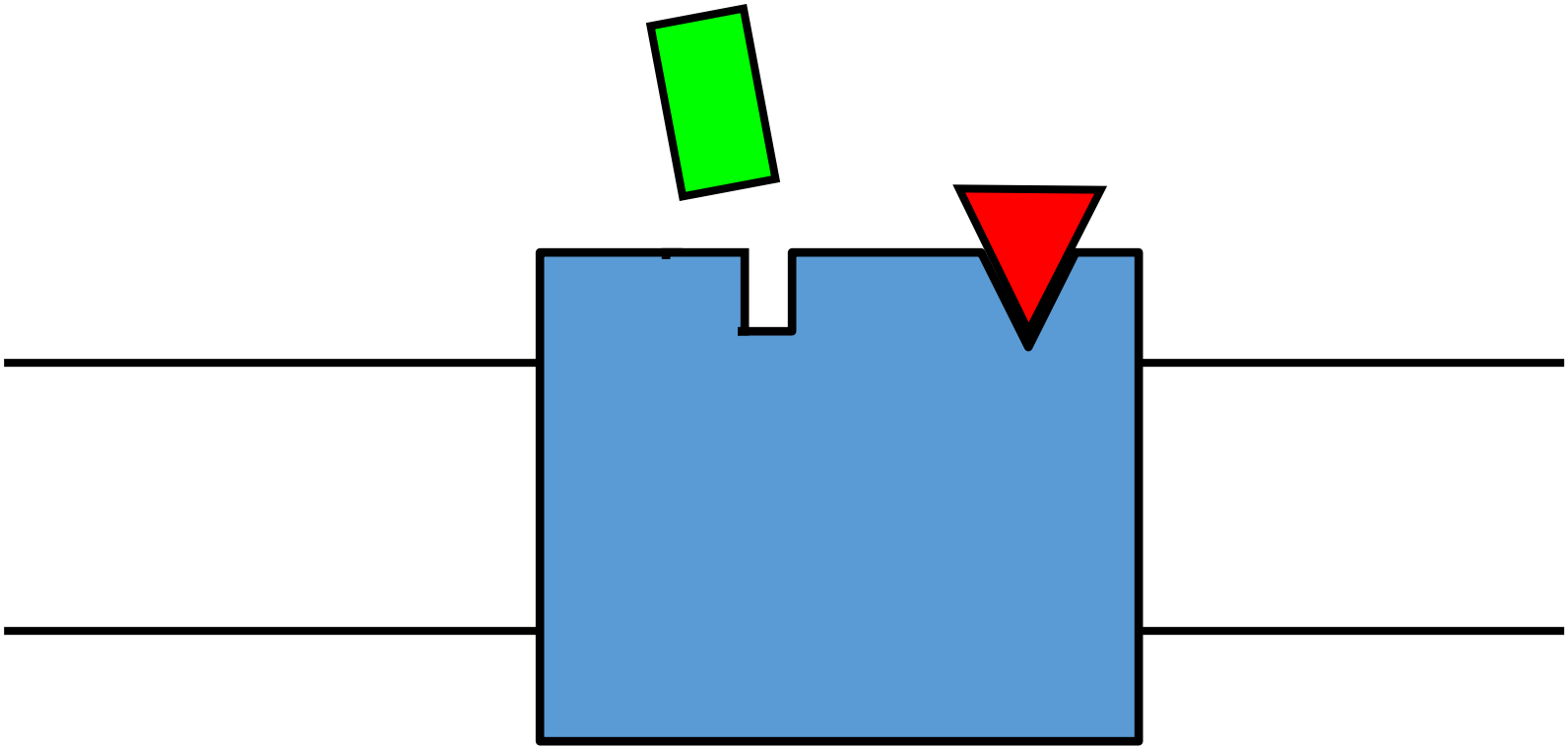
Allosteric Antagonism



Allosteric Antagonism



Allosteric Antagonism



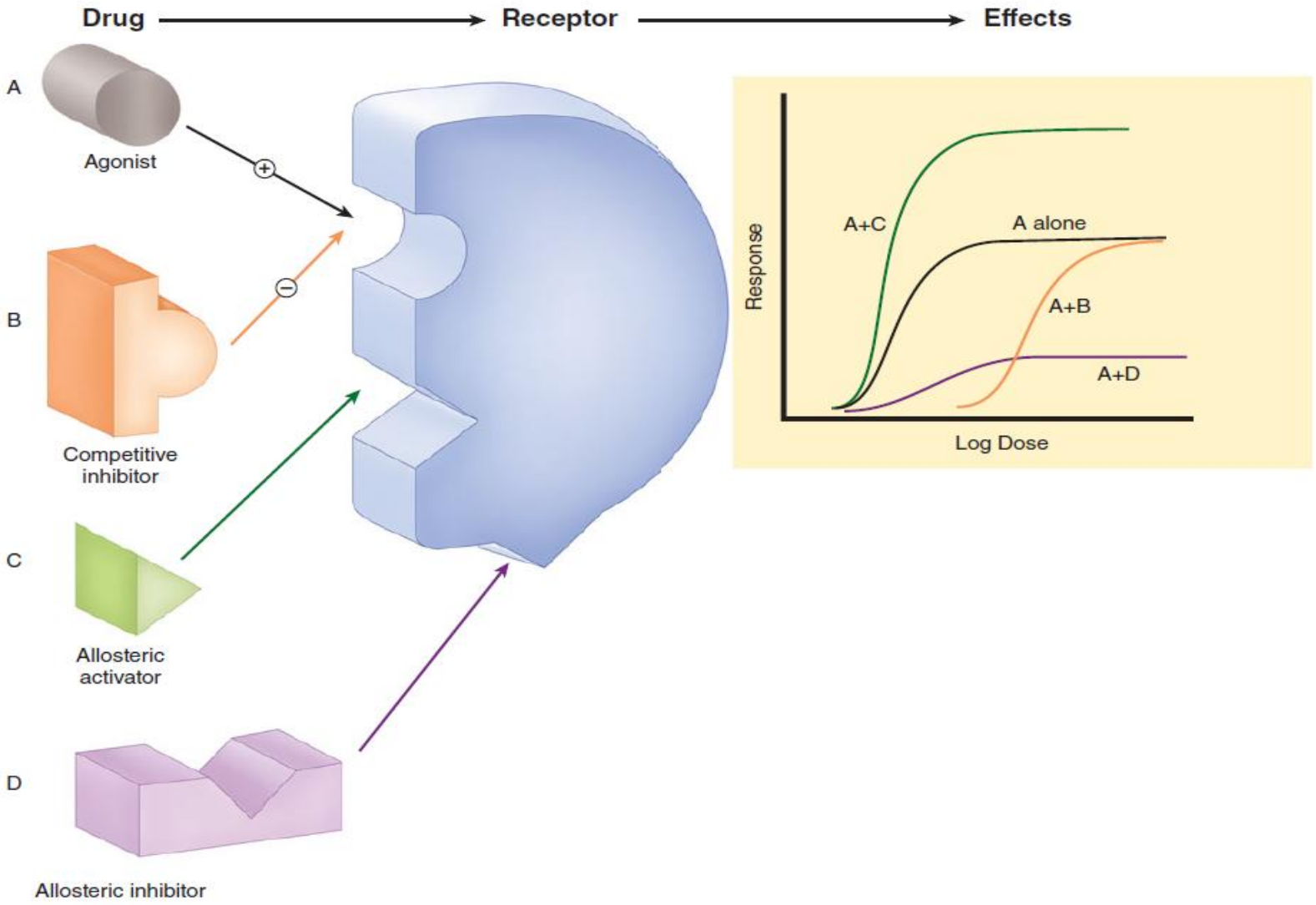


FIGURE 1–3 Drugs may interact with receptors in several ways. The effects resulting from these interactions are diagrammed in the dose-response curves at the right. Drugs that alter the agonist (**A**) response may activate the agonist binding site, compete with the agonist (competitive inhibitors, **B**), or act at separate (allosteric) sites, increasing (**C**) or decreasing (**D**) the response to the agonist. Allosteric activators (**C**) may increase the efficacy of the agonist or its binding affinity. The curve shown reflects an increase in efficacy; an increase in affinity would result in a leftward shift of the curve.

Chemical Vs. Physiological Antagonism

Chemical Antagonism

- **Simple chemical reaction.**
- **No receptor.**

- **<u>Examples:</u>**
  - Heparin & protamine sulfate
  - Iron & Deferoxamine.

Physiological Antagonism

- **Physiological effect is antagonized.**
- **Drugs acting on different receptors:**

- **Examples:**

✓**Norepinephrine → Vasoconstriction → ↑ BP.**

✓**Histamine → Relax vascular smooth muscle→ ↓BP**

Desired vs undesired effects: therapeutic Index , index of drug safety… tolerability profile

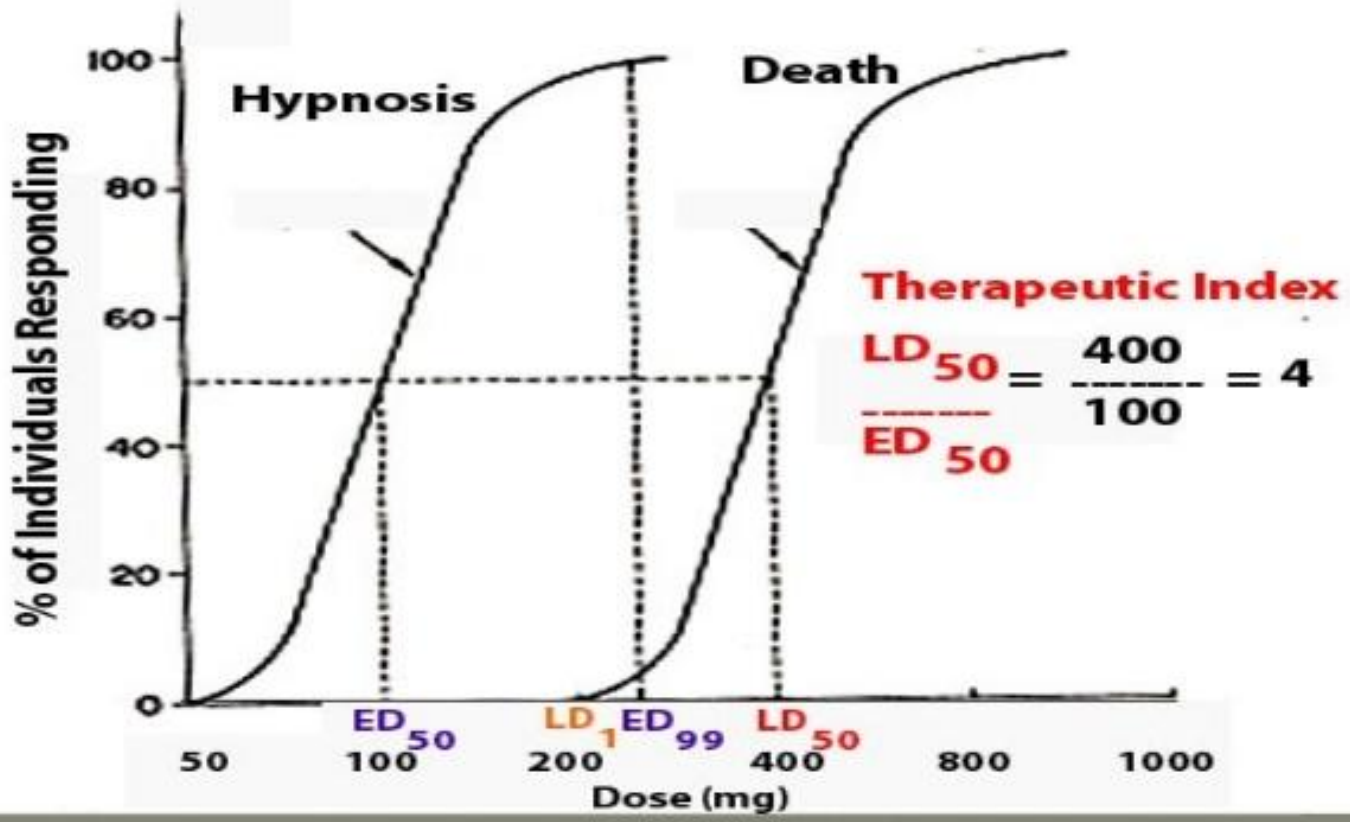
- **ED50 -** Median Effective Dose 50 ; the dose at which 50% of the population or sample manifests a given effect.

- **TD50 -** Median Toxic Dose 50 - dose at which 50% of the population manifests a given toxic effect

- **LD50 -** Median lethal Dose 50 - dose which kills 50% of the subjects

Therapeutic Index = TD50 or LD50/ ED50

Therapeutic Index



Unusual Responses:

Definitions:

1. Idiosyncratic response: unusual response
2. Hyporeactive: less than normal response
3. Hyperreactive: more than normal response
4. Hypersensitivity: allergic or other immunological reaction.
5. Tolerance: decreased response with continued administration
6. Tachyphylaxis: rapidly developing tolerance